The discrete perception of continuous speech

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Abstract

Among the foundational questions for cognition is whether perceptual systems encode sensory input as continuous values or transform it into discrete mental representations. Speech — long a model system for probing the nature of perceptual representation — offers a critical testing ground. Yet the literature routinely conflates the ***content*** of representations (cue-based vs. category-based) with their ***granularity*** (continuous vs. discrete). Likewise, the well-known observation that incrementally varied stimuli often elicit graded behavioral responses, particularly when averaged across trials, is commonly taken as evidence for continuous internal representations. We challenge this assumption and synthesize the surrounding debate by introducing the ***Discrete Sampling*** (DS) framework, a novel theory under which listeners map continuous input into discrete, category-based internal states via a biologically plausible sampling procedure. We formalize a specific instantiation of DS computationally and demonstrate how it accounts for a wide range of core phenomena. We compare DS to a general class of continuous alternatives using data from 575 subjects across three experimental paradigms, with precise, differentiating predictions confirmed by the results. DS provides a single cognitive architecture for a parameter-independent derivation of binary and scalar categorization responses, concave reaction-time patterns centered on category boundaries, the joint modulation of categorization slope and processing speed by task demands and noise, and the distribution of inter-listener variability. We show how sampling from discrete categories can produce behavior traditionally taken to imply continuity, offering a unified reinterpretation of decades of evidence and reframing how uncertainty and apparent gradience are represented in perceptual and psychological systems more broadly.

Keywords: speech perception, Discrete Sampling, perceptual encoding, categorical vs. continuous representations, computational modeling

1 Beyond Identification

How is information from the external world stored in our minds? Whether perceptual encoding preserves the continuity of sensory input (e.g. Getz & Toscano, 2021; Green & Swets, 1966; Mesgarani, Cheung, Johnson, & Chang, 2014; Toscano, McMurray, Dennhardt, & Luck, 2010) or imparts a transformation into inference-supporting mental categories (e.g. Fodor, 1975; Harnad, 1987; Liberman, Harris, Hoffman, & Griffith, 1957) is among the most fundamental — and most elusive — questions in cognitive psychology. In this regard, speech perception is something of a model organism for studying the mental representation of physical variables (Poeppel, Idsardi, & Van Wassenhove, 2008). The tools for precise measurement, manipulation, and synthesis of speech combined with the automaticity of perception allow for unusually rich psycho-physical study, and the robust prediction of the relationship between input audio and resulting behavior. Not only does speech afford top-down influences (Ganong, 1980) — thus serving as a bridge between perception and higher-level cognition — but unlike systems such as vision or memory, our knowledge of speech shows remarkable cross-linguistic variation that must be learned. Speech provides a fertile ground for studying the processes by which our mind encodes this buzzing, blooming world.

Yet regardless of the domain, one must exercise caution in inferring the causal structure of empirical outcomes, particularly when an explanation seems "natural" — intuition can be misleading. This is evident in visual memory: people effortlessly *recognize* objects, yet they are remarkably poor at recalling even the simplest features of highly familiar categories, such as coins or even letters (Nickerson & Adams, 1979; K. Wong, Wadee, Ellenblum, & McCloskey, 2018).

Furthermore, it is an inconvenient truth that markedly different internal architectures can yield highly similar observable behavior. To take a biological example, the camera-style eyes of vertebrates and cephalopods are structurally analogous, both featuring a lens, retina, and adjustable pupil. However, they evolved independently and differ even in their embryonic origins (Nilsson, Johnsen, & Warrant, 2023; Packard, 1972). Similarly, the effect of lexical frequency on processing speed is one of the most robust and well-known behavioral phenomena in psycholinguistics (Forster & Chambers, 1973), yet its definitive cause remains a matter of debate. The effect could reflect either the direct encoding of a continuous measure such as (log) frequency in the mind (Baayen, Milin, & Ramscar, 2016). Alternatively, the same behavioral pattern could simply arise via a serial search on a ranked list of words (Murray & Forster, 2004), an architecture which need not ever encode frequency, even as an intermediate step (Rivest, 1976; Sleator & Tarjan, 1985)[1].

In order to solve this observational equivalence problem — when multiple internal mechanisms suffice to produce the same observable outcome — we need to expand the set of phenomena under consideration. This has been true throughout history: for astronomers in the 16th century, observations of planetary motion could be accounted for either by a geocentric system (with epicycles) or by a heliocentric system; the patterns of motion alone were insufficient to distinguish the two accounts. Only by considering additional evidence, such as the phases of Venus, could the heliocentric system be definitively confirmed (Laudan, 1990).

In the quantitative study of language, one of the most persistent ideas is that properties of the lexicon reflect functional pressures for efficient communication (Piantadosi, 2014; Zipf, 1949, *inter alia*). Yet the supposedly characteristic long-tail distributions of word forms, usage, and meanings would equally emerge under a variety of blind, stochastic processes, often illustrated with analogies such as a "monkey at a keyboard" (Mandelbrot, 1954; Miller, 1954, 1957). Observing the static distribution of words alone is again insufficient to distinguish between these competing accounts. Only by examining additional evidence, like the historical and developmental trajectories of language and acquisition, may we properly assess the relative contributions of communicative pressures versus stochastic forces in shaping the lexicon (Caplan, Kodner, & Yang, 2020).

The challenge of multiple realizability — where simple behavioral patterns underdetermine underlying cognitive architecture — applies as well to gradience versus discreteness in speech perception. Neither the canonical S-shaped identification function (Harnad, 1987; Kuhl, 1991; Liberman

[1] A simple "transposition" algorithm (similar to bubble sort) which stores no item-level counts at all but simply swaps the order of a just-encountered item with the one ranked immediately above it will provably approximate the latent frequency distribution over time. This has numerous practical and applied motivations from situations in which items are accessed repeatedly, but access probabilities are, *a priori*, unknown or impractical to track. This finds a natural home in explaining lexical frequency effects: the speed at which participants respond on a lexical decision is extremely well fit by serial search on a ranked list of words (Murray & Forster, 2004); the procedure by which such a ranked order comes to exist in this form can be easily accomplished via the local transposition algorithm even absent frequency information.

et al., 1957) nor flatter variants observed under different conditions (Apfelbaum, Kutlu, McMurray, & Kapnoula, 2022; Fry, Abramson, Eimas, & Liberman, 1962; Toscano et al., 2010) reveal whether the perceptual system encodes speech along a continuum or via discrete states (McMurray, 2022). A lesson from the analysis of algorithms is useful here: all sorting algorithms compute the same function, but even algorithms with identical input-output mappings may differ wildly in processing time depending on input conditions (Knuth, 1997). We thus draw inspiration from the theory of computation: discriminating among competing theories of speech requires examining "secondary" phenomena that capture the dynamics of processing, such as the concave-downwards[2] reaction-time pattern with a peak at category boundaries (Pisoni & Tash, 1974).

It has long been recognized (e.g., Gerrits & Schouten, 2004; Massaro & Cohen, 1983a; McMurray, Tanenhaus, & Aslin, 2002), including renewed attention in recent work (e.g., Gur & McMurray, 2025; McMurray, 2022; Toscano et al., 2010), that behavioral responses to incrementally varied speech stimuli — steps along a speech continuum for instance — often appear graded, particularly when averaged across trials or participants. This observation is commonly taken to imply that the underlying representations involved are themselves continuous. We challenge that assumption: graded behavioral patterns do not entail gradient representations. Of course, this particular point is hardly new. It has been known at least since the work of Estes (1956); Sidman (1952), one cannot directly infer participant response curves by looking at average response curves. In fact, as Estes (1956, p. 134) puts it, "so also could any given mean curve have arisen from any of an infinite variety of populations of individual curves". Yet, the same work also points out that "Although the form of a group mean curve does not determine the forms of the individual curves, it does provide a means of testing exact hypotheses about them." That is, the average response curve(s) can be used to test specific hypotheses about individual-level response curves, and thus the participants which generate them.

[2]We expand on this reaction-time dynamic at length in the subsection 4.2 but note that while it might informally be described as an "upside-down U shape" or even "parabolic," the latter is not strictly accurate. We instead use the term "concave-downwards" to pick out that the reaction-times have a single slow peak and are monotonically decreasing on either side. As the range of a test continuum grows large then the RT trend should flatten away from the category boundary and is thus not well described by a parabola.

Here we take up this challenge. We introduce a novel framework, and corresponding computational model which instantiates the theory, we call Discrete Sampling (DS). Under DS, the perceptual representation of speech is neither binary (all-or-nothing), nor is it continuous. Rather, listeners construct a perceptual encoding by sampling multiple discrete "labels" from the input stimulus. This encoding allows for the maintenance of uncertainty while listeners' percepts nevertheless exist in one of a countable, finite set of possible states. The DS model is wholly discrete yet explains a wide range of core phenomena in speech processing in a simple, unified way. This includes the ability for listeners to produce either category labels or scalar responses from a single percept, the monotonically decreasing relationship between processing time and distance from the category boundary (Pisoni & Tash, 1974; Rizzi & Bidelman, 2024), and the wide distribution over categorization curve shapes under different experimental paradigms, as well as their specific, joint correlation with task difficulty (Apfelbaum et al., 2022; Gerrits & Schouten, 2004; Kong & Edwards, 2016). Beyond individual trials, we also demonstrate that the DS model accounts for the empirical distribution over apparent inter-subject variability with a single, shared cognitive architecture. All of these phenomena are explained via a system composed entirely out of discrete internal components despite the resulting graded output. Listeners' behavior in response to speech input may be graded, but the internal processes which generates that behavior are discrete.

1.1 Roadmap

The remainder of this article is organized as follows. In Section 2 we review the current literature on the perceptual encoding of speech, drawing an important distinction between the *content* of these representations (whether they encode acoustic-phonetic cues or abstract categories) and their precision or *granularity* (whether they are binary, finite-state, or continuous) — a distinction often conflated in prior discussions. We discuss evidence against both binary representations (an extreme interpretation of "categorical perception") as well as the direct encoding of acoustic-cue information. This leads to a central theoretical contrast between discrete (finite-state) vs. continuous (but still category-grounded) representations in the perceptual encoding of speech. In Section 3 we introduce and fully formalize the Discrete Sampling (DS) framework and model implementation. In Section 4 we demonstrate the DS model's application to and derivation of a range of core phenomena in speech perception, highlighting where novel predictions diverge from gradient theories and the DS model is uniquely supported. Finally, Section 5 concludes with a discussion of what the DS theory does and does not predict, along with broader implications for psychological representations outside of speech.

2 Speech Representations in Two-Dimensions

Despite close to eight decades of study (Cooper, Delattre, Liberman, Borst, & Gerstman, 1952), accounts of speech perception still don't seem to play nice with each other. Many authors emphasize the central role that (linguistic) categories play in perception (Liberman et al., 1957) and mental inference (Quilty-Dunn, Porot, & Mandelbaum, 2023). Yet across the aisle are purely gradient, continuous encoding theories (Goldinger, 1998; McMurray, Tanenhaus, & Aslin, 2009, *inter alia*), which posit that listeners maintain fine-grained acoustic-phonetic detail and even talker-specific information (Goldinger, 1996; Pisoni, 1993) over time. While some authors remain noncommittal to the level of specificity of continuous encoding, others are quite direct: "Listeners are exquisitely sensitive to fine-grained acoustic detail within phonetic categories" (Clayards, Tanenhaus, Aslin, & Jacobs, 2008) and "fine phonetic details about the form and structure of the signal are not lost as a consequence of perceptual analysis [. . .] and apparently become an integral part of the mental representation of spoken words in memory" (Pisoni, 1993).

2.1 Perceptual Encoding (an intermediate representation)

Part of the tension in this debate arises from a lack of clarity about the level of representation under investigation. Is the focus on the contrastive or meaning-bearing representations that form the end state of speech recognition — be it phonological categories or lexical entries — or on the intermediate representations that encode information about the acoustic signal at earlier stages? Sensory input is often ambiguous (Allen & Miller, 1999; Hillenbrand, Getty, Clark, & Wheeler, 1995; Pierrehumbert, 2008), subjected to noise (Johnson, 2004; Mattys, Davis, Bradlow, & Scott, 2012), and dependent on context (A. Samuel, 1996; Warren, 1970). Many kinds of sensory information, including speech, are ephemeral and thus if mental computations are not made over transient and shifting information as it occurs, they cannot be made at all. This leads to a fundamental constraint on perceptual processing termed the *immediacy of computation* (Caplan, 2021; Caplan, Hafri, & Trueswell, 2021; Christiansen & Chater, 2016): systems that extract meaning from sensory signals must encode this information *in the moment* if they are to use it at all. Listeners cannot wait until the end of an utterance to begin building a representation of speech (Allopenna, Magnuson, & Tanenhaus, 1998) and must, in real-time, construct an *intermediate representation* of the input. Here the intermediate representation is a perceptual encoding capable of being held in short-term memory (irrespective of content or form) which outlasts the original input stimulus but may be integrated with additional information over time or used for subsequent decision making. This general process is illustrated in Figure 1. By all accounts, acoustic cues serve as probabilistic evidence for a perceptual encoding, which in turn forms the primary basis of categorization decisions. Nor is it disputed — even under continuous encoding views (e.g. McMurray, 2022) — that listeners can and do *categorize*. Rather the debate must reside in the "hidden layer(s)" (Figure 1); what is the nature of the perceptual encoding that forms the intermediate (pre-output) stage of the recognition process?

Before we can answer that question it serves to mention the generality of the sketch just provided. In the TRACE model of spoken word recognition (McClelland & Elman, 1986), for instance, initial activation of sub-segmental features leads to activation of phones[3], which leads to activation of words that contain those phones. Even after the input to the model is no longer

[3] We use the term "phone" in its broad, descriptive sense to refer to an individual perceptual category corresponding to a speech sound. Linguistic theory often distinguishes between "(allo)phones" — the concrete realizations of speech sounds — from "phonemes" as the contrastive categories differentiating meaning across words. The term "segment" is sometimes used interchangeably with phone to refer to any concretely realized speech unit, though some frameworks treat segments as theoretical abstract units. Our use of phone is intended as a neutral description of sound categories (be they segments, features, or some other abstract categories) rather than making commitments about particular linguistic constructs.

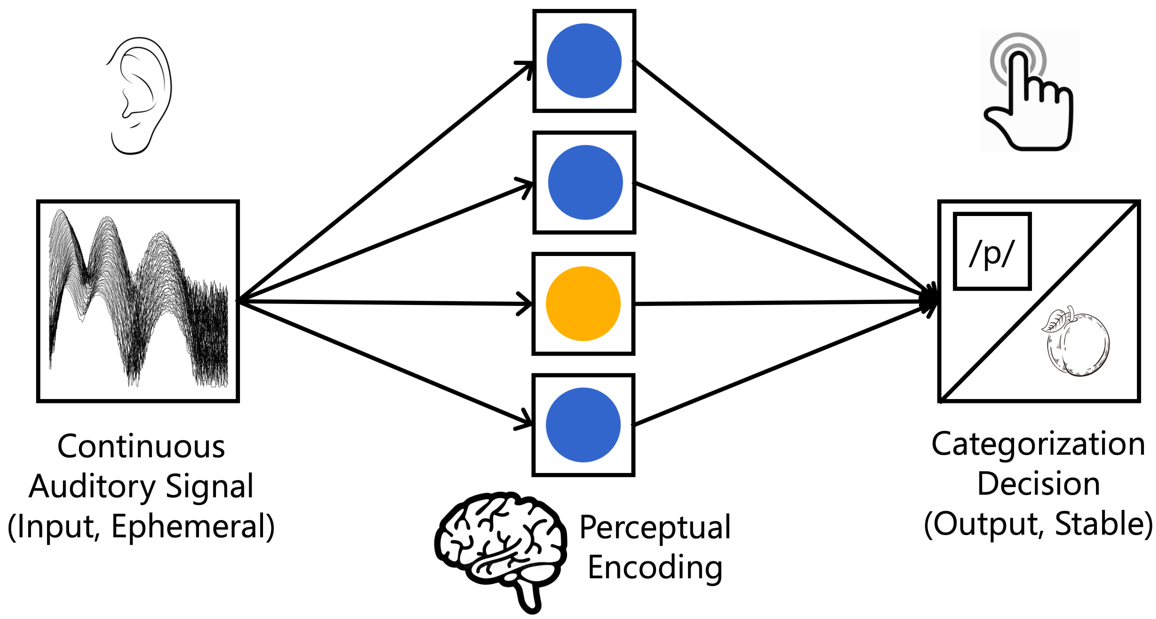

Figure 1: Schematic illustration of the process of perception. The ephemeral input signal must be parsed into a perceptual encoding which serves as the basis of subsequent categorization and decision making.

present, activation of intermediate representations (i.e., phone units or features) persists for some period of time. Importantly, activation cannot "jump" levels: there is no way to activate a word here directly on the basis of sub-phonemic features without first activating the intermediate stage of phones. Listeners cannot directly map from acoustic input to the highest-level resultant categories; reaching the "end-state" of perception must be accomplished through some intermediate perceptual encoding of the input.

In TRACE, for instance, this activation is not "all or nothing" and allows for sensitivity to graded differences in the input (Allopenna et al., 1998; McMurray et al., 2002) — although as we discuss throughout this article, sensitivity to graded differences in the input does not entail gradient representations themselves. Note that the phone units in TRACE represent only one possible type of intermediate unit that listeners may use to recognize speech; any intermediate representation that is abstracted away from the initial input and allows for graded activation can potentially serve as a relevant representation. This intermediate structure serves as a listener's working hypothesis for recognition, but given the immediacy of computation constraint, the form and content of these representations are a bottleneck: computation can occur only over the material that is constructed and maintained rather than the original, ephemeral signal. What's more, since information continues to arrive in the input stream, it is important for the mechanisms involved in even the earliest stages of processing to pass information up to higher levels of representation that are able to support subsequent inference.

The present article is concerned primarily with this perceptual encoding: the format of information that listeners construct directly from the original, ephemeral speech signal, prior to a relatively stable or strict commitment to an abstract category. This transformation of the initial input into an intermediate perceptual encoding is of particular importance since, given the immediacy of computation (Caplan et al., 2021; Christiansen & Chater, 2016), downstream processes can only operate over what is preserved in this representation. Thus the content and granularity of the intermediate encoding imposes a bottleneck on every subsequent stage of processing.

2.2 Terminology

Another issue endemic to the existing literature is that terms which dominate discussion are often defined only loosely, and used inconsistently across studies (Lilienfeld, Pydych, Lynn, Latzman, & Waldman, 2017; McMurray, 2022; Moore & Skidmore, 2019, etc.).

This is perhaps typified by the phrase "categorical perception" (CP); which has shaped theorizing about the psychology of language and the mind for decades (see e.g., Goldstone & Hendrickson, 2010; Harnad, 1987; McMurray, 2022). Categories are clearly indispensable for perception (e.g. Best, McRoberts, & Goodell, 2001; Feldman, Griffiths, & Morgan, 2009; Kleinschmidt & Jaeger, 2015; Kuhl, 1991). However, what was originally described as the "extreme assumption" — henceforth "extreme-CP" — dating back to Liberman et al. (1957) that listeners commit to the interpretation of a single category interpretation at the earliest stage of processing, with no retention of between-category uncertainty is untenable in light of widespread evidence to the contrary. Even though within-category sensitivity on discrimination tasks is weaker than between-category discrimination, listeners are consistently above chance at differentiating tokens that fall within the same phone category (e.g. Fry et al., 1962; Pisoni, 1971). There is a range of well-attested, top-down effects in speech perception (e.g. Ganong, 1980); in order for phonetically ambiguous stimuli to be nudged towards the category which forms a real word (e.g. "dash" vs. *"tash") then across-category uncertainty must persist at least long enough to interact with lexical context. There are also so-called "long-distance cue integration" effects where listeners combine information from temporally separated cues before committing to a category (e.g. Connine, Blasko, & Hall, 1991). If all-or-nothing categorization were immediate, then later-arriving cues should either have no effect or they should completely override the initial phonetically-driven decision; the fact that temporally disjoint information displays an interaction means that it must be modulated by some constructed and stored uncertainty.

Leaving aside the fact that Liberman et al. (1957) never, in fact, claimed that the psychological function generating their observed discrimination data was extreme-CP (see Appendix A), listeners clearly must construct and maintain some kind of uncertainty about the perceptual categories in question, both in real-time comprehension (Connine et al., 1991) and in subsequent memory and learning (Caplan et al., 2021; Norris, McQueen, & Cutler, 2003; A. G. Samuel & Kraljic, 2009). At the same time, and as we discuss here, it would be equally misleading to equate the graded behavioral outcomes observed in these tasks with a fundamentally continuous underlying representation. Thus, the original spirit of categorical perception — the critical role of categories in shaping perceptual experience — remains important, even if its classical formulation requires substantial revision.

2.3 Content and Granularity of Encoding

Any terminological inconsistencies notwithstanding, the largest conceptual issue at play is the conflation of two distinct dimensions on which representations can vary. The first dimension concerns the *content* of the representation: whether it is based in acoustic-phonetic ***cues*** or abstract ***categories***.

This pair of terms is frequently invoked but rarely defined together in a consistent way. For our purposes ***cues*** are measurable properties of the acoustic signal. They provide probabilistic evidence for higher-level phonetic or linguistic distinctions (e.g. VOT[4] is a measurable cue, whereas [$\pm$voicing] or /t/-/d/ are categories). Cues are objective properties of the input signal itself and thus exist separate from the observer and independent of linguistic structure. Conversely, while ***categories*** may be inferred from cues (as well as top-down information), they represent a fundamentally different kind of information. Categories are language-specific *referential units* that function as elements within larger compositional structures such as syllables or words. Thus the perceptual encoding of speech could, in principle, contain cue-based or category-based information.

The second dimension concerns the *granularity*, or precision, of the representation: how many bits of information would be required to store any particular value. This could be binary (an all-or-nothing representation, i.e. a single bit), discrete (allowing a finite number of unique states across a few bits), or truly gradient/continuous in which case the number of distinguishable mental states is effectively unbounded. We expand on this second dimension further below, but granularity is a property which could equally apply to either cue-based or category-based encodings. This allows us to define a complete 2×3 (6-cell) table of how the possible theories of the perceptual encoding of speech fit into the space of content *vs.* granularity (Figure 2).

While many classic (and contemporary) studies involving categorical perception have equated category-based with binary representations, these dimensions are orthogonal.[5] They are logically independent of one another. A listener may encode category-level information in a graded or continuous format, or alternatively maintain cue-based acoustic information in a discretized format (e.g. low-, medium-, or high-VOT). Such possibilities have been implicit in several lines of work — for instance, McMurray, Aslin, and Toscano (2009); Toscano and McMurray (2010) emphasize how graded acoustic evidence informs discrete perceptual and categorization decisions.

[4]"voice onset time" (VOT) is the time delay between the release of a stop consonant and the onset of glottal pulses from the closed vocal folds. VOT is the primary acoustic cue for distinguishing voiced stops (e.g., /b/, /d/, and /g/) from their voiceless counterparts (/p/, /t/, and /k/).

[5]The proposal here is certainly not the only possible distinction one could draw, but we are committed to the idea that *some* distinction is required. Du and Durvasula (2025), while proposing a slightly different set of distinctions, argue for a similar need to disentangle the relevant dimensions conflated in theories of representation in Phonology and Speech Production

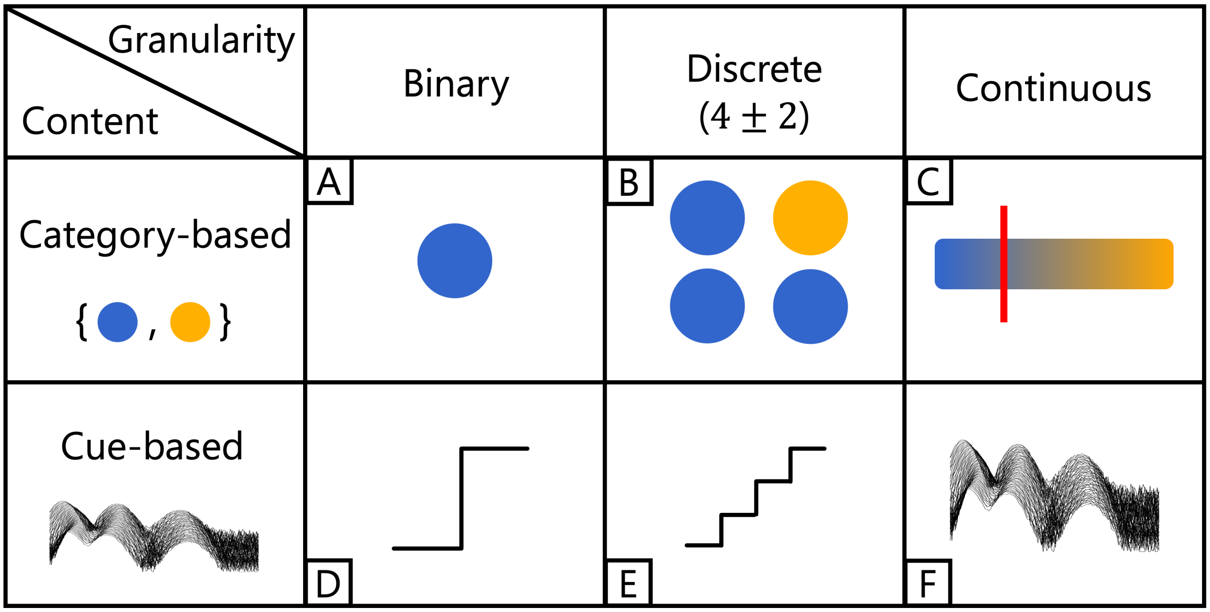

Figure 2: Two-way classification of theories of the perceptual encoding of speech based on content (category-based vs. cue-based) and granularity (binary, discrete, continuous). These dimensions are theoretically independent of one another but have been frequently conflated in the literature. "Cues" are measurable, objective properties of the input signal like voice-onset time (VOT). "Categories" (e.g. [±voicing] or /t/-/d/) are language-specific referential units taking part in compositional structure.

Both reviews (Harnad, 1987) and potential explanations of CP-behavior (Kuhl, 1991; Liberman & Mattingly, 1985) note that categorical behavioral outcomes need not imply categorical representations (and vice versa with respect to gradient behavior vs. representations). Nevertheless, binarity and category-based theories have been coupled more tightly than is warranted; by making the two dimensions of granularity and content explicit, we aim to clarify the full space of theoretical possibilities. In doing so, we note both substantial convergence across seemingly opposed traditions but also reframe the central unresolved question: whether graded behavior has its basis in truly continuous encoding or a limited space of discrete representations.

To build some intuition around granularity, take a simple analogy at the interface of vision and engineering: high-speed vs. low-speed video cameras. These may be distinguished from each other on the basis of their shutter-speed (or frame rate) which, together with pixel resolution, defines the composition of the resulting video. Shutter-speed is a physical constraint. While we can manufacture cameras with higher or lower sampling rates, ultimately each device carries some limit. The effect of increasing the shutter-speed, which increases granularity of the resulting video and thus the bits required to store that video, is precisely an increase in the precision of any particular representation. Later on — Section 3 — we use the symbol N within the Discrete Sampling framework to represent this physical constraint in speech perception, but this should be understood as shorthand for this sort of physical property rather than a free parameter of the model.

A digital video camera is not truly continuous of course, but apparent continuity arises due to stitching together a sufficiently large number of still frames. If we take a video of a lion running in the savanna, a human observer watching this playback can use the encoded frames to make a decision (e.g. is this a lion or a cheetah?). If the shutter-speed is high enough then any minor changes to it may not have an external effect — being overshadowed by other properties like lighting quality or top-down information like our knowledge of savanna animals generally — and the frame rate may appear effectively unbounded. Though under the right conditions (the stroboscopic "wagon-wheel" effect for instance (Purves, Paydarfar, & Andrews, 1996)), this inherent physical constraint becomes apparent once again. Whether each snapshot records raw-signal (a mega-pixel image) or purely higher-level information ("does this image contain stripes?") a greater shutter-speed means a more granular encoding of the original input.

Thus "granularity" provides a way to describe the number of mental units which comprise a given mental encoding. Under extreme-CP we have a binary alternation over possible categories (cell A), thus listeners would be storing exactly one piece of information: a single label indicating the end phone in question. Rather than an all-or-nothing perceptual encoding, we could still build a finite and discrete representation (i.e. a set of 4 ± 2 labels, itself still limited, countable[6]), or to construct a truly continuous representation (in which case the number of distinguishable internal states is effectively unbounded).

While the evidence against the extreme-CP view — as above, listeners must clearly construct

[6]This approximate range of 4 ± 2 is not an arbitrary restriction and can be viewed as in line with the literature on short-term memory (Cowan, 2001; Miller, 1956), just-noticeable differences (Pisoni, 1977), as well as subitizing (Atkinson, Campbell, & Francis, 1976) though we provide a more specific derivation of this later when we fully specify the DS model, section 3.

and maintain some kind of uncertainty during the perceptual process — is widespread (Connine et al., 1991; Ganong, 1980; Norris et al., 2003, *inter alia*), this simply rules out the binary hypothesis. Thus we reject the extreme-CP view under which listeners could only manipulate a single mental variable; and each act of perception would result in an all-or-nothing decision about a category label. Yet we should not let the pendulum swing too far in the other direction. To impose the constraint that listeners possess the ability to store non-determinism (uncertainty) in a perceptual encoding in no way entails that such a representation reflects arbitrary precision, or actual probability information. Indeed, to reject extreme-CP should not be an endorsement of or even evidence in favor of either a cue-based or category-based representation, nor does it privilege a theory which allows for true gradience (arbitrary granularity/precision) rather than a discrete number of mental states ($\approx\ 4 \pm 2$). Perceptual uncertainty removes the "Binary" cells from consideration but this still leaves us with a 2×2 theoretical space: encodings could be cue- or category-based, and could be encoded in either continuous (granularity $\rightarrow \infty$) or discrete (4 ± 2) format. We discuss each of these distinct combinations of content (cue vs. category) and format (continuous vs. discrete) in turn.

Continuous-cue theories (Cell F) assume that perceptual representations preserve fine-grained acoustic-phonetic detail (e.g., $VOT = 18ms$, $F_1 = 650Hz$, etc.) stored and used in a continuous format. These models treat the perceptual encoding as essentially a rich cue vector or distribution over vectors, postponing categorical commitment until task demands require it. As described in McMurray, Tanenhaus, and Aslin (2009, p. 87): "fine-grained acoustic detail is preserved (in gradient, not discrete form) for at least several hundred milliseconds. . . during lexical access, the system is sensitive to fine-grained detail, and this detail is retained throughout processing." We will argue that the effect such authors describe actually reflect uncertainty represented via discrete states but the theoretical stance is clear. Connectionist models such as TRACE (McClelland & Elman, 1986) as well as exemplar and episodic accounts (Crowder & Morton, 1969; Goldinger, 1998; Goldinger & Azuma, 2004; Johnson, 2004; Pierrehumbert, 2008) also fall squarely in the fine-grained camp.

By contrast, Discrete-cue theories (Cell E) preserve the same status of cues as representational primitives, but assume that their values are stored only coarsely, quantized into a small set of discrete states (e.g., VOT or F1 $= \{low, medium, high\}$). Such a "quantized cues" view suggests that the perceptual system trades precision for robustness, encoding cue values (and thus sub-categorical phonetic detail) in bounded steps rather than with arbitrary granularity. Quantal theories of speech (e.g. Blumstein & Stevens, 1979; K. N. Stevens, 1989; K. N. Stevens & Keyser, 2010) provide a classic rationale for this kind of coarse-grained cue-encoding, arguing that perceptual categories arise from stable acoustic "regions" created by nonlinearities in the articulatory-acoustic mapping — an idea that has found a home in work on second-language (L2) perception (e.g. Iverson & Evans, 2007).

Shifting from cues to categories (which we provide evidence in favor of below, see Section 2.4), Continuous-category theories (Cell C) propose that the representational primitives are *categories* — phonological or lexical hypotheses — but their activation is a scalar property. At any given moment, the system encodes a probability or continuous activation over possible categories (e.g. /b/: 0.72, /p/: 0.24, /other/: 0.04). This is the view posited by Bayesian (Norris & McQueen, 2008) or ideal-observer models of recognition (Bicknell, Bushong, Tanenhaus, & Jaeger, 2025; Kronrod, Coppess, & Feldman, 2016) which frame speech perception as a problem of probabilistic inference over categories. While some authors here use the term "subcategorical," this work really seems to be describing gradience ***about*** or ***across*** categories rather than ***within*** categories (and perhaps the term "supracategorical" would be more apt); so this seems most accurately to fit in Cell C. Notably, even work focused on the limits of cue maintenance has seemed to endorse continuous representations over categories (Caplan et al., 2021), illustrating the logical separability of representational content from format.

Finally, Discrete-category theories (Cell B) occupy the remaining quadrant. Here representations are still fundamentally *about* categories, but uncertainty is represented in a bounded, finite fashion — for instance via a small set of possible states, (e.g., definitely-/b/, probably-/b/, ambiguous-/b∼p/, probably-/p/, definitely-/p/). Uncertainty is thus represented not as a continuous probability distribution but as a bounded set of labeled samples drawn from the category space. The framework we introduce in this article — Discrete Sampling — is an instantiation of this approach, demonstrating how perceptual uncertainty (and resulting graded behavior) can be represented in discrete terms without requiring continuous representations.

2.4 Evidence against cues

How can we adjudicate between the four remaining possibilities (Discrete/Continuous and Category-/Cue-based) given that, as outlined in Section 1, simple categorization functions suffer from an observational equivalence problem? One striking line of evidence comes from perceptual learning. When listeners encounter target words in which an acoustic cue to a phone is systematically manipulated, their category boundary subsequently shifts (e.g. Norris et al., 2003; A. G. Samuel & Kraljic, 2009) — listeners have adapted to this particular "accent." This may be naturally induced by embedding locally ambiguous /t/-/d/ segments in word/nonword pairs like "croco[t/d]ile." Listeners may use lexical context (similar to the Ganong effect (Ganong, 1980)) to resolve the ambiguity: only one interpretation of the input results in a real word, allowing listeners to infer the intended category (/d/). The perceptual learning effect is that this interpretation generalizes to future contexts *without* lexical support — the interpretations of steps along a simple /ta/-/da/ continuum are affected by prior exposure to the "croco[t/d]ile"-type stimuli. These kinds of lexically guided adaptation effects are quite easy to induce in learners, requiring only minimal input (one or two dozen target exposures), and attested across a range of tasks with different demands including word counting, syntax, or loudness judgments (Drouin & Theodore, 2018; McQueen, Norris, & Cutler, 2006).

Yet this sort of lexically-guided perceptual learning is equivocal with respect to whether the perceptual encodings required to support it contain cue-based or category-based information. Since the ambiguity on the exposure trials in such cases can be resolved *immediately*, perceptual learning can be accomplished regardless of listeners' ability to encode and maintain cue-based information. That is, if lexical content serves to activate the target phonological category *before* the ambiguous audio arrives, then listeners can adapt to systemic changes even without maintaining acoustic cue information over time; lexical context alone suffices to guide categorization. Under a cue-based encoding theory, this specific timing — that disambiguation occurs prior to the target audio — is not particularly load-bearing. That is only, of course, if listeners can maintain signal-dependent information in a representation over time. On a category-based encoding theory, however, the ***timing*** of disambiguation thus plays quite a critical role in supporting perceptual learning: if the correct linguistic category were not active before the ambiguous audio then there would be no information to generalize and thus perceptual learning could not occur.

Caplan et al. (2021) provide a stringent test along exactly these lines. The Caplan et al. paradigm employs word-onset minimal pairs (e.g., “time” vs. “dime”). This way no word-internal information will provide the correct interpretation of a locally ambiguous segment (/t/ vs. /d/). The paradigm further manipulates the timing of disambiguating information via orthographic subtitles: text occurs either slightly before or slightly after the onset of audio. When text follows acoustic input, perceptual learning requires retention of acoustic-cue information. Under a purely category-based perceptual representation, the system would encode uncertainty over something abstract and discrete like “/t/ vs /d/” or [$\pm$voicing] (Durvasula & Nelson, 2018), but the cue-specific evidence would be lost once categorization occurs, leaving no basis for learning. This is the precise empirical pattern found: perceptual learning only occurs when disambiguating information precedes the ambiguous input (Caplan et al., 2021). This pattern provides evidence against both continuous- and discrete-cue representations (i.e. everything in the “Cue-based” row of Figure 2). Even a coarse, quantized-cue representation (e.g., $VOT \in \{low, medium, high\}$ rather than millisecond-level values) would have been sufficient to support perceptual learning — listeners would just need to maintain “medium-VOT” and then map it to either /d/ or /t/ depending on condition. The in-principle exposure mapping from “medium-VOT” to one of the two speech categories was 100% deterministic within an experimental condition, yet perceptual learning does not occur when disambiguation follows the onset of acoustic input.

We note that the empirical finding from Caplan et al. (2021) does not stand in isolation. The attested timing asymmetry in perceptual learning is robust even at extremely compressed durations: Jesse and McQueen (2011) report that Dutch listeners exhibit perceptual learning from ambiguous target phones which appear word-medially ("bene[f/s]it") or word-finally ("regre[ss/ff]") but not word-initially ("[f/s]reedom") — see also McAuliffe and Babel (2015) for a similar word-position manipulation in English fricatives. This follows if listeners are able to use local context to recognize or predict the containing lexical item by the point that the word-medial or word-final ambiguous segment appears (P. A. Luce, 1986) but not in the word-initial condition. The timing differences between Jesse and McQueen's as well as McAuliffe and Babel's word-initial and word-medial lexical disambiguation conditions may be as small as a couple-hundred milliseconds; suggesting a lack of cue-based representation (at least which can be recruited for downstream inference like perceptual learning) even at markedly early timescales — though we return to discussion of perceptual encoding at extremely early stages in the General Discussion (Section 5). The perceptual learning timing asymmetry thus replicates across labs and stimulus editing techniques, across languages (at least in Dutch and English), both within- and across-word boundaries, across disambiguation cues (orthography vs. lexical context), and across phonological contrasts (fricative place vs. stop voicing). What's more, there is empirical evidence that lexically-guided perceptual learning generalizes to new phonetic contexts (Nelson & Durvasula, 2021); suggesting that the adaptation effect is neither driven by nor located in specific acoustic-cue values.

Previous work has shown that listener behavior is sensitive to within-category phonetic differences (Brown-Schmidt & Toscano, 2017; Falandays, Brown-Schmidt, & Toscano, 2020) and that listeners may integrate disparate sources of information over a temporal delay (Galle, Klein-Packard, Schreiber, & McMurray, 2019; Gwilliams, Linzen, Poeppel, & Marantz, 2018). However, such findings are primarily evidence against a binary encoding (the single bit case) and do not reveal the internal format of the representations which support this sensitivity. Falandays et al. (2020), for instance, exposed listeners to ambiguous audio edited along a continuum between /h/ and /ʃ/, which in turn could manifest as referential uncertainty over the pronouns ("he" vs. "she") used to identify Mickey and Minnie Mouse. Using an eye-tracked visual-world paradigm participants indicated whether a spoken sentence matched a display. The key manipulation was the length of the temporal gap between ambiguous audio and disambiguating context. In the longest condition, participants maintained uncertainty between the two referents over the course of 35 intervening syllables. We view this finding as something of a *reductio ad absurdum* for the cue-based approach. Either listeners are storing cue-based uncertainty indefinitely (itself difficult to square with the perceptual learning literature, e.g. Caplan et al. (2021); Jesse and McQueen (2011)) or this uncertainty is accomplished via a category-based representation. These categories are as high-level as discourse referents in the case of Falandays et al. (2020).

While the received wisdom of the field has often implicitly assumed that rejecting extreme-CP automatically favors some unbounded, continuous representation (i.e. $\neg A \therefore C$), our exposition thus far has shown that this inference is unwarranted, motivating a more precise investigation of the discrete vs. continuous nature of category-based representations. Ruling out binary encodings, and cue-based encodings leaves us with a much more specific question and the central issue under investigation: are the category-based perceptual encodings that comprise the perceptual encoding of speech built in a continuous (arbitrary precision, cell C) or discrete (4 ± 2, cell B) format? In order to answer this question we next (Section 3) introduce the Discrete Sampling (DS) model as a concrete formalization and implementation of a cell B theory. This will be used, in turn, to contrast empirical predictions against continuous representation alternatives and we demonstrate that DS is uniquely supported.

3 Discrete Sampling Model

We introduce a novel theoretical framework and computational model for the perception of continuous speech using wholly discrete internal states: the Discrete Sampling (DS) theory. DS instantiates a Discrete-Category theory (the top-middle cell in Figure 2), in which internal primitives are *categories*, and uncertainty is represented in a bounded, finite fashion via a small set of possible states. The DS model is fully formalized and its open-source implementation is available for free research (re)use.[7]

Speech perception under DS proceeds in two stages. In the first stage, auditory information is **sampled** as it occurs to form a perceptual encoding[8] (Figure 3). This intermediate representation (the perceptual encoding) is critical: the original acoustic signal is inherently ephemeral, and any subsequent computation or decision must operate over the information that has been extracted and stored, utilizing the format in which it is represented (Caplan, 2021; Christiansen & Chater, 2016). Each sample corresponds to a single category label, and the full perceptual encoding consists of a multiset of N independent labels. Importantly, this sampling process is a biologically plausible strategy for representing transient audio: even if an ideal observer could compute exact likelihoods over continuous input, biological/neural systems are computationally constrained (Gallistel & King, 2011; Sanborn & Chater, 2016). Sampling provides a tractable mechanism by which the brain can represent uncertainty (Hoover, Sonderegger, Piantadosi, & O'Donnell, 2023) and perform inference over a finite set of possibilities — without requiring intractable calculations over a lossless continuous distribution.

In the second stage, the perceptual encoding is used to generate task-relevant responses. Depending on the experimental paradigm, this may involve inferring a single phonemic or lexical category (Ganong, 1980; Liberman et al., 1957; McMurray et al., 2002) or reporting a continuous-scale estimate of category-likeness (Massaro & Cohen, 1983a; Pisoni & Tash, 1974). The DS model formalizes how these behavioral outcomes can be derived directly from a discrete internal representation through a principled resampling procedure (Appendix C.2-C.3) for handling binary and continuous output respectively.

[7]Source code is available on GitHub (https://github.com/scaplan/marbles) as well as the Open Science Framework — OSF (https://osf.io/vqr4u/).

[8]A note to the reader: this (i.e. "perceptual encoding") and other key terms used throughout are defined together, for reference, in the Glossary (Appendix D)

3.1 Perceptual encoding

We start with the initial perceptual encoding of input. Each phonological category is taken to have a general, shared boundary μ (e.g. $\sim$30ms VOT for the case of stop voicing). The value reflects an empirical property of perception that can be estimated in behavioral experiments, rather than a free parameter of the model. To capture moment-to-moment psychophysical variability, the DS model evaluates each sample with respect to a category boundary drawn from a Gaussian distribution with mean μ (the "true" boundary) and standard deviation σ. This Gaussian function follows from both neural and psychophysical evidence. σ must be non-zero since neural responses are inherently noisy (Stein, Gossen, & Jones, 2005), even across repeated presentations of identical input (Kanitscheider, Coen-Cagli, & Pouget, 2015; Leonard, Baud, Sjerps, & Chang, 2016). Synthesizing the responses of multiple sensory neurons (each its own small source of variability, e.g. Romo, Hernández, Zainos, and Salinas (2003)) must, by the central limit theorem, yield an approximately Gaussian distribution over the decision threshold applied across samples or trials[9]. At the behavioral level, psychophysical discrimination tasks consistently reveal error patterns well described by Gaussian distributions (Green & Swets, 1966; Hautus, Macmillan, & Creelman, 2021), supporting the idea that perceptual boundaries fluctuate stochastically (σ) around a stable central value (μ).

Each auditory input is represented as a value along a primary phonetic cue dimension (e.g., VOT for a /t/-/d/ contrast, or F_1 for an /u/-/a/ contrast). To generate the perceptual encoding on a given trial, listeners sample from the input and apply the category boundary to each sample independently: if the input is less than the sampled threshold, the label is assigned to one category (e.g., /d/); if greater, to the other category (e.g., /t/). Repeating this process N times produces a multiset of labels that constitutes the perceptual encoding for that trial (see Figure 3A and Appendix C.1), where N can be thought of as an implementation of the "shutter speed" from our earlier camera analogy — or multiple simultaneous cameras each taking a snapshot. N is simply the DS formalization of the physical constraint on granularity.

[9] Alternatively, the distribution could be log-normal (Lyon, 2014), though this distinction is not germane for present purposes.

The role of variability is worth mentioning here. If $\sigma = 0$ that would mean all perception would be completely deterministic, in such a case it would not be possible for a listener to generate a "mixed" perceptual encoding with both /t/ and /d/ labels simultaneously; such a case would devolve into the "extreme" view of categorical perception. Conversely, as σ grows arbitrarily large, it would lead to a correspondingly flat distribution over the probability of potentially phonological categories perceived (independent of the phonetic input). A reasonable value for σ for any particular problem must be estimated empirically though; we expect that it should vary depending on the particular task and context. Differing paradigms and inputs naturally impose different degrees of difficulty, noise, or working-memory load, so we should expect σ to increase or decrease accordingly as a function of task or stimulus properties — with corresponding effects on both perceptual encoding and behavior (we expand on this precise point in Section 4.4).

3.2 Mapping from encoding to behavior

Unlike ephemeral acoustic input, a perceptual encoding is a relatively persistent mental representation. It can be integrated with other information over time and used to guide behavioral responses. For tasks requiring a single discrete output (e.g., phone identification), a multiset of N labels alone is insufficient; it must be mapped onto a category judgment. An analogous situation arises in visual perception. Early visual processes, such as edge detection, yield an intermediate representational format that captures the geometric structure of the input but does not, by itself, specify object identity (*a la* the primal sketch; Geisler, 2008; Marr, 1982). Further transformations are required to derive an identified object representation. Drawing inspiration from evidence accumulation for perceptual decision making (e.g. K.-F. Wong & Wang, 2006), in DS this is achieved by a **re-sampling** procedure: listeners repeatedly draw labels (with replacement) from their perceptual encoding until they obtain M consecutive identical labels (e.g. $M = 5$ would require five draws of a /d/ in a row until a /d/-response can be output). At that point, the corresponding category label is fixed, and behavior is executed. This procedure (Figure 3B and Appendix C.2) not only produces trial-level responses but also naturally provides a measure of reaction time (RT) to deci-

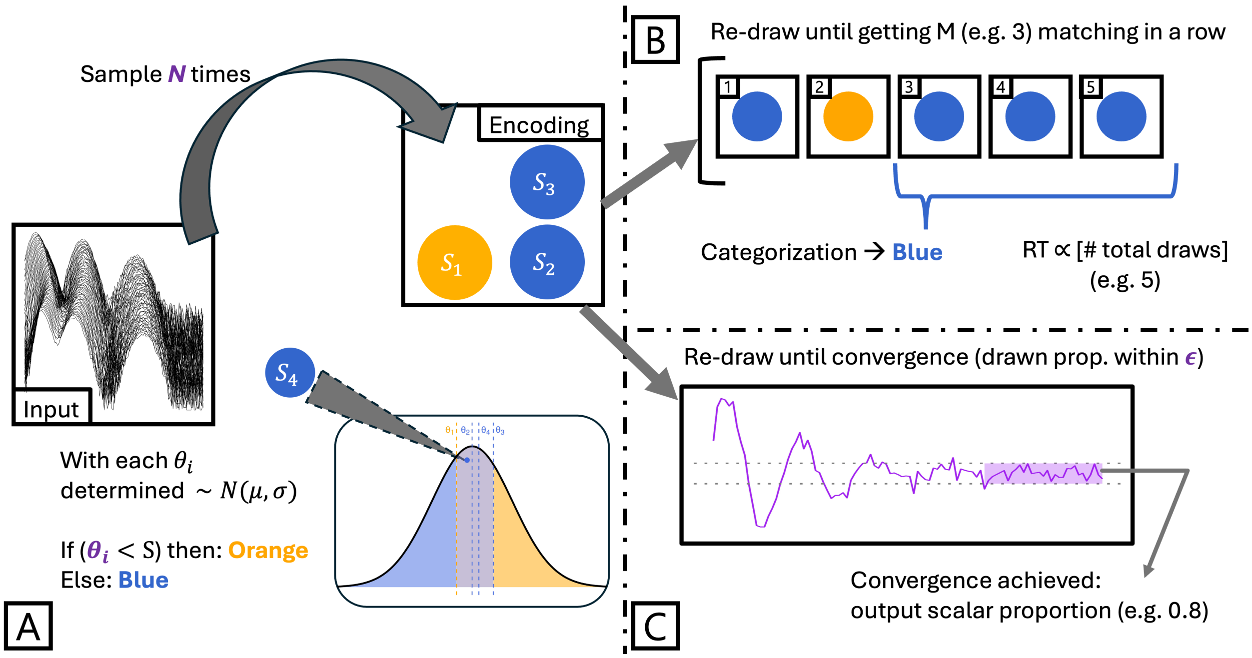

Figure 3: Schematic illustration of the core procedures in the Discrete Sampling (DS) model. Panel A shows the procedure to sample a single perceptual encoding. $input$ is a primary acoustic cue such as F_0 or VOT which is sampled N times. mu is the general category boundary subject to $sigma$ variability. The process to generate a single category judgment from a perceptual encoding is shown in Panel B: the encoding of N labels is drawn from (with replacement) until M consecutive identical labels have been obtained and a category output produced. The number of draws to decision is a natural measure of reaction time (RT). Panel C shows the analogous process to generate a continuous scalar response from an encoding: labels are sampled in a buffer of M draws until convergence on the proportion of samples is achieved. Algorithmic pseudocode for all of these processes is provided in Appendix C.

sion, based on the number of draws required to reach a run of M consecutive matched labels. This re-sampling procedure is a biologically plausible mechanism for mapping a discrete perceptual encoding onto a behavioral response (Gold & Shadlen, 2007; Ratcliff & Rouder, 1998; K.-F. Wong & Wang, 2006). The brain cannot directly compute an exact posterior over the relevant categories, and maintenance of phonological uncertainty from the input must be mediated through the perceptual encoding. The number of draws (i.e. rate of evidence accumulation towards a decision) required to reach M consecutive identical labels naturally indexes trial-level uncertainty: when the perceptual encoding is strongly skewed towards one category, convergence occurs quickly. As the encoding reflects a state of greater uncertainty, more draws are required to resolve the ambiguity. This provides a natural link between behavior, processing time, and the entropy of the underlying perceptual representation.

For tasks involving graded or continuous responses (e.g., VAS or Likert-scale judgments of /t/-likeness), the same re-sampling logic applies, but requires a slightly amended procedure (see Figure 3C and Appendix C.3). Instead of stopping after M consecutive matches, the procedure continues until a sufficient number of draws has been collected to generate a stable estimate of the category proportion. The reported scalar is the proportion of drawn labels corresponding to the target category. This approach naturally accounts for differences in RT across the perceptual continuum: convergence occurs quickly when the perceptual encoding is strongly biased toward a single category (low-entropy regions) and more slowly near ambiguous inputs (high-entropy regions). Such behavior aligns with psychophysical observations showing longer RTs near category boundaries (Pisoni & Tash, 1974; Rizzi & Bidelman, 2024), providing a principled link between discrete internal states and graded human responses.

3.3 Evidence for the number of discrete states

We stress that the DS model is not a curve-fitting exercise. The central boundary μ is an empirical property. Likewise, the boundary noise σ reflects the inherent stochasticity of perceptual thresholds, which has long been measured in psychophysical paradigms as "sensitivity" (i.e., the slope of a psychometric function). Lastly, the number of labels N in the perceptual encoding reflects a biologically constrained sampling capacity, not an arbitrary parameter to be tuned. In practice, model predictions are robust across a broad range of N (see Section 4); even extremely limited encodings ($N \in \{2, 3\}$) are sufficient to capture the graded response and RT patterns observed in human listeners as a function of phonetic input. This insensitivity to N beyond $N \geq 2$ highlights an important theoretical point: the perception of continuous speech does not require gradient representations, rather a limited number of discrete states which encode uncertainty, generated under natural physical variability, are sufficient to explain a wide-range of empirical phenomena via a discrete sampling procedure. Below we provide three lines of evidence motivating why such limited values for N ($\sim 2-6$) are *a priori* the most reasonable when considering known neurological and behavioral constraints.

The parameter N should be interpreted as the number of *separable* and *independent* samples that can be or are extracted by the auditory cortex in the brief window during which this information is available. Work on oscillatory speech coding places strict limits on this temporal resolution: while the auditory nerve can phase-lock at very high frequencies, temporal precision declines substantially in cortex, with characteristic timescales consistent with sampling on the order of tens of milliseconds (Beach et al., 2021; Giraud & Poeppel, 2012; Oderbolz, Poeppel, & Meyer, 2025; Poeppel, 2003)[10]. On our conservative reading of this literature, this places cortical temporal resolution in the neighborhood of 20-50 ms per meaningful "frame," implying roughly $\sim 2-6$ independent samples for a typical phone (assuming a duration of 100-200 ms).

Psychophysical constraints are further in line with this general window of small-N. Just-noticeable-different (JND) estimates for primary phonetic cues like VOT show that listeners cannot reliably discriminate between arbitrarily small temporal offsets — with JNDs commonly in the range of $\sim 5-10$ms across phone boundaries and $\sim 20-30$ms within-phones (e.g., Aslin, Pisoni, Hennessy, & Perey, 1981; Pisoni, 1977). Using such JNDs as a guide, this would divide a common range of VOTs tested in an experimental continuum (e.g. 0-80ms) into just 4-8 distinct buckets. The same is true for other cues: the JND for distinguishing fricatives like /s/-/ʃ/ is approximately a 20% change in the spectral frequency. The perceptual system need not — and in fact cannot — directly represent a continuum at millisecond or Hz precision across many independent samples. Taken together, these neural and psychophysical constraints provide a principled *a priori* range for N at around $2-6$ samples.

Perhaps most intuitively, we note that the approximate range of 4 ± 2 can be viewed as following the literature on subitizing — our subconscious ability for automatic and accurate extraction of numerosities of visual objects without explicit counting or serial enumeration (Atkinson et al., 1976). Subitizing seems to function at precisely the range of 4 ± 2 (Trick & Pylyshyn, 1994).

[10] In addition to the 20–50 Hz sampling window associated with phoneme-level processing, the literature also identifies slower cortical rhythms in the 3–6 Hz range which appear to support syllable- or prosody-level parsing of speech (Giraud & Poeppel, 2012; Poeppel & Assaneo, 2020). The former is primarily relevant for this article.

This aligns, as well, with the number of items we can typically hold in short-term memory before requiring some hierarchical or chunking strategy (Cowan, 2001; Miller, 1956).

Similarly M can best be understood as the overall speed or deliberateness for generating behavior in response to auditory input on a trial. Yet, while a larger M correspondingly increases the number of calculations to decision (slower RTs) and a smaller M has the opposite effect (faster RTs), the RT *dynamics* are unaffected by M: independent of specific values, encodings are mapped to behavior (whether category judgments or continuous responses) in the same way and with the same characteristic concave-downwards RT curve. The parameter M formalizes how much information the system reads out from the perceptual encoding to commit to a decision, but as with the size of the sampled encoding N, this does not qualitatively affect the results in an interpretively meaningful way.

The combination of N and M set the granularity of the perceptual system's internal state space. To solve a recognition problem involving K phonemic categories — $K = 2$ for a binary contrast or $K \approx 11$ for British English monophthongs (Williams & Escudero, 2014) — the number of distinct perceptual encodings grows combinatorially with N (Equation (1)), as does the number of "bits" of memory required to represent a point in that space logarithmically (Equation (2)).

$$\text{States}(N, K) = \binom{K + N - 1}{K - 1} \tag{1}$$

$$B(N, K) = \log_2\big(\text{States}(N, K)\big) \tag{2}$$

Even at modest values of N, the DS model admits a large, yet finite, state space sufficient to account for graded behavioral responses to input without requiring the impractically large precision required to support truly continuous representations. The further implications of a tight bound on N become clear when we consider how the representational state space scales. Even modest increases in N (for recognition problems of K phone categories) quickly yield an explosion in the number of possible states (Table 1). For a binary contrast, encodings with $2 \leq N \leq 6$ require only a handful of states (≈2 bits). But for an 11-way vowel contrast (like British English monophthongs), raising N to something approximating gradience, say 100, implies more than 46 trillion possible states (>45 bits), a value far beyond what psychophysical constraints could support.

K-way Phonemic Contrast	N sampled labels in Perceptual Encoding	#Possible Distinct States	# Bits
2	1	2	1.000
2	2	3	1.585
2	3	4	2.000
2	5	6	2.585
2	100	101	6.658
3	1	3	1.585
3	2	6	2.585
3	3	10	3.322
3	5	21	4.392
3	100	5,151	12.331
5	1	5	2.322
5	2	15	3.907
5	3	35	5.129
5	5	126	6.977
5	100	4.6 Million	22.133
11	1	11	3.459
11	2	66	6.044
11	3	286	8.160
11	5	3,003	11.552
11	100	46.8 Trillion	45.415

Table 1: Number of possible distinct perceptual states as a function of the number of sampled labels in the perceptual encoding (N) and the size of the phonemic contrast set (K). The final column shows the information-theoretic capacity in bits required to represent each possible state.

Taken together, these scaling considerations underscore the necessity of a principled implementation. With small values of N, DS naturally yields a compact set of discrete internal states that remains tractable for both neural and cognitive systems, while still capturing the variability and graded sensitivity to input observed in human behavior. To move beyond intuition, it is useful to formalize DS mathematically: to define precisely what a perceptual encoding comprises, and how internal components of the model govern both the generation of such perceptual encodings and their application in mapping onto behavior. Section 3.4 provides this formalization, completing the foundation for quantitative predictions and empirical evaluation of the DS model as a unified explanation for the discrete perception of continuous speech.

3.4 Formalization of the DS framework

Here we formalize the DS model before describing its application to empirical data. Let $\mathcal{P}$ denote the set of phones relevant to the particular perceptual problem at hand (whether in the listener's entire language or just for a specific task), for instance:

$$\mathcal{P} = \{/b/, /d/, /g/\}$$

Within the Discrete Sampling framework, a specific perceptual encoding e is formally represented as an (unordered) multiset of size N of "labels" sampled from $\mathcal{P}$. This is defined by a function that maps phone labels to their count (i.e. multiplicity):

$$e : \mathcal{P} \to \mathbb{N}_0, \quad \text{such that} \sum_{\varphi \in \mathcal{P}} e(\varphi) = N.$$

The value $e(\varphi)$ gives the multiplicity of phone φ in the encoding e, and N is the cardinality of e — the total count of all sampled phone labels in the encoding. $\mathbb{N}_0$ is simply the natural numbers including 0.

Each such e can be interpreted as a *distinct* perceptual representation composed of N labels (ignoring order but allowing repetition of course). The set of *all* such multisets — the finite space of possible perceptual encodings constructed from N sampled labels over the phone set $\mathcal{P}$ — is likewise defined as:

$$\mathcal{E}_N(\mathcal{P}) = \left\{ e : \mathcal{P} \to \mathbb{N}_0 \;\middle|\; \sum_{\varphi \in \mathcal{P}} e(\varphi) = N \right\}.$$

If the number of relevant phonemes in the language is $|\mathcal{P}| = k$, then the size of this space of possible perceptual representations is given by the number of multisets (encodings) of cardinality N on k phones:

$$|\mathcal{E}_N(\mathcal{P})| = \binom{N + k - 1}{k - 1}.$$

The model/listener's perceptual encoding is a single element e drawn from this space $\mathcal{E}_N(\mathcal{P})$ according to a probability distribution which we derive below (Section 3.4.1) as determined by a phonetic input and the relevant category threshold.

3.4.1 Probability of a specific encoding

The probability of a particular encoding e resulting when the listener receives some phonetic input x (e.g. a VOT value) can be derived as follows. Consider, for simplicity, a binary phone set $\mathcal{P} = \{/t/, /d/\}$ and an encoding with N labels. For each label i, its value is determined by a threshold θ_i, with each applied category boundary drawn independently and identically from a normal distribution[11]:

$$\text{label}_i = \begin{cases} /t/ & \theta_i \le x \\ /d/ & \theta_i > x \end{cases}, \quad \theta_i \sim \mathcal{N}(\mu, \sigma^2).$$

First, we find the probability p that a single, arbitrary sample (label) is categorized as /t/, given a phonetic input x and the applied threshold $\theta_i \sim \mathcal{N}(\mu, \sigma^2)$. In order to help with an intuitive understanding of the formalization, we provide a visualization of the categorization in Figure 4, where higher values correspond to category /t/ and lower values to /d/. The figure shows the distribution of values that the category boundary can take. It also shows a value taken by the category boundary in a specific instance ($\theta_i = \mu = 30$) and a specific input ($x = 35$).

From the figure, it is easy to see that an input x will be categorized as /t/ whenever $\theta_i \le x$. Therefore, the probability of categorizing a specific input x as the category /t/ is equivalent to the probability that the category boundary is less than or equal to the input value.

$$p = P(\text{label}_i = /t/ \mid x) = P(\theta \le x)$$

Since $\theta_i \sim \mathcal{N}(\mu, \sigma^2)$, we standardize this probability:

[11] Here we assume that an input is labeled as the higher category when $\theta_i = x$; given that the variable is continuous however, there is no there is no meaningful effect of this assumption.

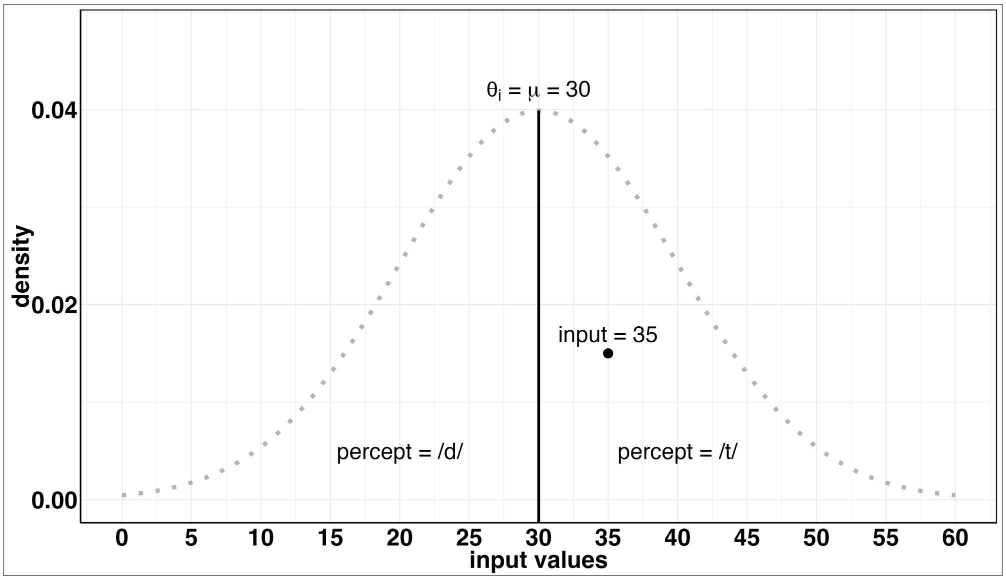

Figure 4: A visualization of the categorization process for a single labeled sample under DS.

$$p = P\left(Z < \frac{x-\mu}{\sigma}\right) = \Phi\left(\frac{x-\mu}{\sigma}\right)$$

where Z is a Gaussian variable and Φ is the standard Gaussian cumulative distribution function (CDF). The probability of a single sample (label) being categorized as /d/ is the complement:

$$P(\text{sample} = /d/ \mid x) = P(\theta > x) = 1 - p = 1 - \Phi\left(\frac{x-\mu}{\sigma}\right)$$

Since an encoding e is defined by its counts, for our binary case this is:

$$e(/t/) = n_t, \quad e(/d/) = N - n_t, \quad \text{where } 0 \leq n_t \leq N.$$

Intuitively, the probability of obtaining this *specific* encoding e is the probability of sampling exactly n /t/ labels and $N - n$ /d/ labels. Because the N applied thresholds $\{\theta_1, \theta_2, \ldots, \theta_N\}$ are independent, the resulting labels are in turn independent. Let $e(/t/) = n_t$ and $e(/t/) = n_d$ be the respective counts (multiplicities), with $n_t + n_d = N$. The number of /t/ labels n_t therefore follows a Binomial distribution:

$$n_t \sim \text{Binomial}(N, p)$$

The probability of the specific encoding e is given by the Binomial probability mass function. The final result is:

$$P(e \mid x) = \binom{N}{n_t} p^{n_t} (1-p)^{N-n_t}, \quad p = P(\theta \leq x) = \Phi\Big(\frac{x-\mu}{\sigma}\Big),$$

It is further possible to prove that the expected value of the number of computations required map from an encoding e to output behavior will be greatest when e holds a uniform distribution over the category labels in question. However the proof does not provide any intuition (and thus we elide it for space/clarity) beyond the idea that if all labels in an encoding come from a single category, the representation is deterministic in generating behavior: M matching labels will be re-sampled in exactly M steps (the fastest possible response). If labels are uniformly distributed among alternatives in e then the uncertainty of any given draw is highest, and long sequences of matching draws are least predictable (the slowest possible responses on average). We return to this point, along with accompanying simulations in Section 4.2.

3.4.2 A concrete example

We are now able to work out a complete example with specific phonetic values and a category boundary. Let x (a VOT value) be `27`, let the size of perceptual encodings N be `3`, and let the category boundary mean μ be `30` with variance σ^2 `2`. We can work out the probability that, for instance, the listener's encoding e contains exactly two /d/s and one /t/ as follows, i.e. $P(e = \{= /d/ : 2, /t/ : 1\} \quad | \quad 27)$. The probability p that a single sample is categorized as /d/ is:

$$P(\text{/d/} \mid x) = P(\theta_i > x) = 1 - \Phi\left(\frac{x-\mu}{\sigma}\right)$$

Substituting the values $x = 27$, $\mu = 30$, and $\sigma = 2$:

$$p = 1 - \Phi\left(\frac{27-30}{2}\right) = 1 - \Phi(-1.5) \approx 1 - 0.0668 = 0.9332$$

The probability of the encoding e is given by the binomial probability mass function (with the probability distribution over perceptual encodings is shown in Table 2):

$$P(e \mid x = 27) = \binom{3}{2} p^2 (1-p)^1 = \binom{3}{2} (0.9332)^2 (0.0668)$$

$$P(e \mid x = 27) = 3 \times 0.8709 \times 0.0668 \approx 3 \times 0.05818 \approx 0.175$$

e	n_d	n_t	$P(e \mid x = 27)$
$\{/d/, /d/, /d/\}$	3	0	0.8129
$\{/d/, /d/, /t/\}$	2	1	0.1745
$\{/d/, /t/, /t/\}$	1	2	0.0125
$\{/t/, /t/, /t/\}$	0	3	0.0003

Table 2: Probability distribution over all possible perceptual encodings for an input VOT of $x =$ 27ms, given a category boundary $\theta_i \sim \mathcal{N}(30, 4)$ and $N = 3$ samples.

Having introduced the DS framework and worked through a concrete example of how discrete perceptual encodings give rise to graded output behavior, we now turn to the model's specific predictions. The DS model not only accounts for but *explains* a wide range of core findings in speech perception. Importantly, while a very natural interpretation of the DS model is as a middle-ground between $N = 1$ extreme-CP and ($N \to \infty$) gradience, Discrete Sampling is not a hybrid or compromise between discrete and continuous systems[12]. It is fully discrete in structure, with listeners able to encode information only via a finite, countable number of internal states, yet it naturally produces graded *output* behavior, explains the particular relationships we see between task difficulty categorization slope and reaction times, as well as variable behavior slopes across tasks and participants. As we show below, these phenomena result from the qualitative *structure* of Discrete Sampling rather than any particular quantitative values.

[12]Such a "compromise" theory for instance may posit that the perceptual encoding simultaneously comprises *both* category- and cue-based information, as in Figure B1, or contains both discrete and continuous information

4 Results

We are now ready to evaluate the predictions of the Discrete Sampling (DS) model against empirical data in speech perception. Across five case studies, we demonstrate that a small number of discrete states to encode certainty vs. uncertainty across perception is not only sufficient to derive general trends characteristic of human speech perception, but also yields novel, testable predictions which are further confirmed in the behavioral data from over five hundred participants across three experimental paradigms. We proceed as follows: first, we show that category judgments under DS across phonetic inputs are qualitatively independent of the number of sampled labels (N) in the perceptual encoding (Section 4.1); second, we demonstrate that DS provides a natural, and remarkably precise, explanation of reaction-time curves tied to perceptual uncertainty while ruling out several alternative views which fail to account for the RT data (Section 4.2); third, we extend this reaction-time analysis to graded response tasks such as visual-analog scales (VAS) and confirm several very specific novel predictions which results from DS (Section 4.3); fourth, we consider how differing task demands and noise shape categorization output across different experimental paradigms as consistent with DS (Section 4.4); fifth, and finally, moving from individual phonetic inputs and tasks to analyses *across* sets of human subjects we show that the attested distribution of inter-participant variability is actually consistent with and predicted by the DS model under a single cognitive architecture (Section 4.5); highlighting the DS framework's explanatory power for both aggregate trends as well as trial-, subject-, and population-level behavior.

4.1 Category judgments are robust to encoding size

One initial question to consider is whether the size of the perceptual encoding (N) exerts a strong influence on categorization behavior. If DS required large values of N to derive empirical response patterns it would begin to approximate a gradient theory. If it required task-specific tuning of N, it would undermine DS as a general explanatory framework. Conversely, if categorization behavior under DS remains stable across our *a priori* expected small values of N, this would tell us that discrete representations are sufficient to derive the characteristic response curves for speech perception tasks and also that the parameter-independence of behavior with respect to N allows us to use a plausible value for encoding size (e.g. $N = 5$) for future tests without needing to re-fit N individually. We note that N is intended as a fixed value that may (eventually) be estimated empirically rather than a free-parameter to be tuned in the model.

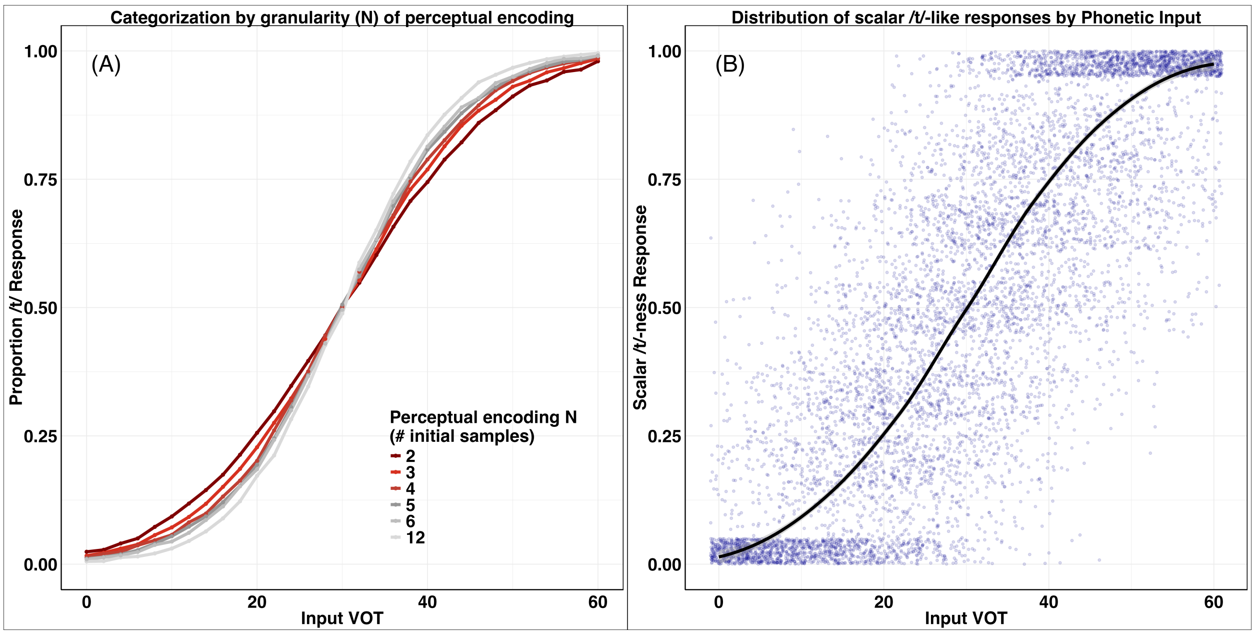

Figure 5: Left panel (A): output of the DS model producing category judgments using different values of N to control the size of perceptual encodings. Right panel (B): output of the DS model used to map perceptual encodings to a scalar-response. Each blue dot is a single response per trial; the black line shows group average. Both tests use the same parameters: category boundary $\mu = 30$ and variation $\sigma = 15$, input tested between 0-60ms at increments of 2ms with 10 trials per VOT level and 1,000 simulated participants.

To test the behavior of the DS model in this regard we performed a large number of simulations according to the following configuration. Treating this as a stop-voicing (/t/-/d/) recognition problem, we set the phonetic input to represent VOT with a mean category boundary μ of 30ms (Lisker & Abramson, 1964; Nakai & Scobbie, 2016) and $\sigma = 15$; the re-sampling/evidence accumulation threshold $M = 5$; input tested between 0-60ms at increments of 2ms with 10 trials per VOT level and 1,000 simulated participants. We simulated categorization judgments in the model while varying N (perceptual encoding size) across six different values between 2 and 12 samples. The categorization output of DS is remarkably stable as a function of N (see Figure 5A): the number of internal samples taken from the phonetic input does not have a meaningful effect on the resulting categorization output — confirmed statistically via nested mixed-effects logistic-regression model comparison: a model with an additional effect of perceptual encoding size N did not significantly outperform the same model with no effect of N ($\chi^2(1) = 1.08$, $p = .30$). As we show in Section 4.4 categorization slopes under DS primarily reflect perceptual sensitivity due to task demands rather than arbitrary memory capacity.

As has received attention in the literature recently (e.g. McMurray, 2022), the particular shape of the response function (even at the subject-level) is not informative as to the gradience or discreteness of the underlying representations at play. This is an averaging issue: even if N were reduced to 1 — akin to implementing the extreme view of categorical perception — this would still derive a smooth cline due to calculating an average across trials so long as there is *any* variation in the application of the category boundary applied (i.e. $\sigma > 0.0$). This issue, famously discussed by Massaro and Cohen (1983a), has been known in the literature since at least 30 years prior (Estes, 1956; Sidman, 1952). This is made clear by a simple analogy: imagine 100 filament light bulbs arranged beneath a luminance meter (holding voltage/current fixed). Suppose we wished to reverse-engineer the internal properties of individual bulbs but, like in our perception experiments, we cannot look inside the bulb and we are measuring the group of individuals as a set. If we track the overall output over time by the luminance meter we'll see a smooth-clined sigmoid function of decreasing brightness. What does this tell us? That each light bulb emits a scalar degree of light that gradually decreases? We, of course, know this is not the case; individual bulbs are either functioning or not. When the filament breaks each bulb no longer produces any light at all; their underlying behavior is categorical, with possible states $\{\text{on}, \text{off}\}$. But since there is variation over the *time* at which each individual bulb may burn out (either due to manufacturing conditions or natural variation as analogous to σ), the resulting aggregate curve of brightness obscures the nature of the underlying process.

One strategy to circumvent this averaging issue has been to use continuous rating tasks (e.g., visual-analog scales — VAS) rather than forced-choice categorization (Apfelbaum et al., 2022; Massaro & Cohen, 1983a); for instance, tasking participants with rating how /t/-like vs. /d/-like a given input stimulus is. Yet this, too, is potentially problematic: if perceptual encodings are discrete, participants may reinterpret the scalar question in ways that still yield graded responses, much as they do when reasoning over concepts which are clearly and necessarily categorical. When supplied only a continuous rating scale, participants judge "7" to somehow be a better odd-number than "57" (Armstrong, Gleitman, & Gleitman, 1983).

Regardless of the prior theoretical stance one may take on the most ecologically valid paradigm for studying perception, the DS model provides a principled account of the situation. Because responses are generated by re-sampling from a discrete perceptual encoding, the same architecture can yield either binary judgments or continuous ratings depending on task demands (i.e. via the Algorithms in Figure 3B-C; Appendix C.2-C.3 respectively). As shown in Figure 5B, even with a small encoding size ($N = 5$) the model produces smoothly graded scalar outputs at the trial level using the same configuration as with categorical judgment output ($\mu = 30, \sigma = 15, M = 5, N = 5$) — again tested on input from 0-60ms in increments of 2ms across 1,000 simulated participants.

Taken together, these results highlight two properties. First, both binary category judgments and graded response patterns emerge naturally from discrete encodings under DS, and independently of encoding size N, providing a promising framework for understanding patterns in speech perception across paradigms. Second, we note that curve shape alone — whether in categorization or continuous rating tasks — does not adjudicate between gradient and discrete representations. To better distinguish accounts we must turn to a more sensitive diagnostic: the reaction-time dynamics of perceptual judgments.

4.2 Concave reaction times reflect uncertainty in perceptual encodings

The preceding results reflect the challenge of multiple realizability; how participants' categorization behavior is affected by shifting phonetic input would seem to be the most direct reflection of the system in question yet it underdetermines the cognitive architecture which generates that behavior. Such a situation is not unique to speech perception and we can draw inspiration for solving this problem from successes where they have occurred in other areas. Consider the analysis of algorithms in computer science. All properly implemented sorting functions compute the exact same mapping from input to output — each will take a list of comparable objects, like integers, in any order and return a list in one-to-one correspondence with the input but whose elements are in sorted order. In fact, there are an *infinitely many* programs which compute the exact same function (Rogers Jr, 1987), each particular algorithm is really a member of an equivalence class that solves the same computational problem.

Yet hope remains; returning to the sorting function example, we can reverse-engineer which algorithm is being used to solve a given problem since implementations may vary considerably in other observable characteristic like run time complexity. For example, the merge-sort algorithm requires a fixed logarithmic time — $O(n \log n)$ — regardless of the internal properties of the to-be-sorted input, while the insertion-sort algorithm runs in linear time — $O(n)$ — for incidentally already-sorted input, but in quadratic time — $O(n^2)$ — for reverse-sorted input (see Goodrich, Tamassia, and Mount (2011) for general content on the analysis of algorithms). Even if running time is not, in and of itself, as central a concern as strict correctness of a particular implementation, this is a powerful tool of inquiry in reverse-engineering. A substantial slice of cognitive psychology may be framed as reverse-engineering problems.

Here we apply a similar analysis to the time dynamics of speech perception as a function of different inputs. A classic, though perhaps underappreciated, aspect of perception is that participant reaction times to stimuli are not uniform across any given phonetic continuum but display a characteristic concave downward (upside-down U-shaped) curve with a peak at the category boundary. Listeners are slowest to respond when the input is at the point of maximal ambiguity and faster to respond as the input moves away from the boundary in either direction. This finding dates back to Pisoni and Tash (1974) whose classic data is shown in Figure 6A. Participant responses to auditory input along a /d/-/t/ continuum were about 90ms ($\sim$15-20%) slower at the category boundary relative to comparable audio further from the category boundary. We confirmed the generality and

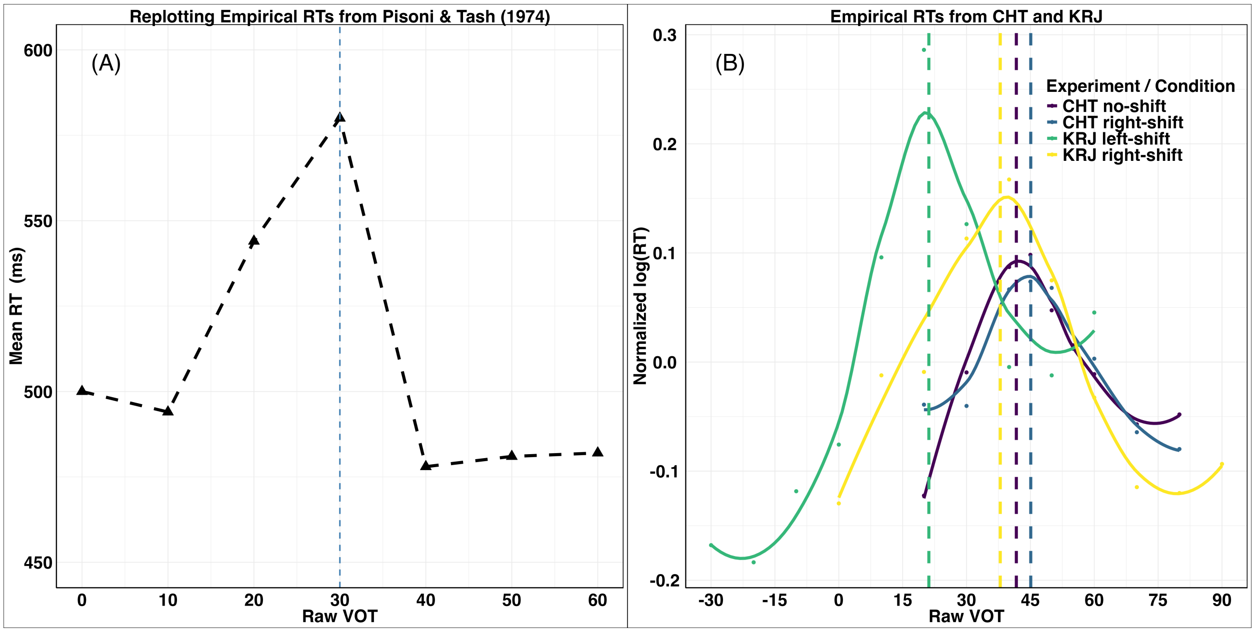

Figure 6: The concave downward relationship between reaction time and phonetic input with a peak of the slowest responses at the category boundary. On the left (A) is data replotted from Pisoni and Tash (1974); the X-axis is input VOT while the Y-axis shows reaction time in milliseconds. The category boundary on the same stimuli is indicated in the dashed blue line at 30ms. On the right (B) is a replication of the same concave RT curve effect in both Kleinschmidt et al. (2015) and Caplan et al. (2021); X-axis is input VOT and the Y-axis is normalized (z-scored within each experiment/condition) log-RT to better compare across experiments. Vertical dashed lines indicate the category boundary for each experiment or condition.

robustness of this finding by investigating the presence of the concave RT trend in a pair of contemporary perceptual and distributional learning experiments: Caplan et al. (2021) used a subtitle-controlled perceptual learning paradigm to shift categorization boundaries tested on a subsequent /d/-/t/ continuum, while Kleinschmidt et al. (2015) used a repeat exposure distributional learning paradigm to shift categorization along the /b/-/p/ continuum up or down. Across the board, we see the same concave RT trend with the slow-peak anchored to the category boundary (see Figure 6B). This appears to be an emergent property of the speech perception process as it does not depend on the precise nature of the input (these experiments used different stimuli, speech continua, and editing procedures) or on a task designed to capture this RT relationship. The paradigm in Pisoni and Tash (1974) was an identification task using purely synthesized speech (Glace, 1968), the data from Caplan et al. (2021) come from the test-phase following perceptual-learning exposure using stimuli splicing together original speech across both VOT and pitch (F_0) cues, while the output from Kleinschmidt et al. (2015) is on trials throughout the extended distributional learning phase. This RT curve is thus not an artifact of some particular audio file or splicing technique since we see the same trend across both natural and synthesized stimuli using different editing procedures and different contrasts.

One might consider the possibility that the RT slowdown is a by-product of lifelong frequency of exposure (*a la* the Perceptual Magnet effect (Guenther & Gjaja, 1996; Lacerda, Karlsson, & Lindblom, 1995)) rather than the categorization or perceptual process since listeners are likely to have encountered more instances of high or low VOTs with the fewest instances of VOT around 30ms say (Allen & Miller, 1999). Yet this cannot be the cause since as in Figure 6B the peak of the RT curve correspondingly shifts up or down, even for the exact same stimulus, depending on whether the distributional learning task has pushed the category boundary up or down (Kleinschmidt et al., 2015). This in-tandem change indicates that the slowdown is a direct function of the category boundary; the RT trend seems to track the state of maximal uncertainty with strong precision.

This RT curve is derived naturally under the discrete sampling model. Recall the procedure for mapping from a perceptual encoding to a category judgment (Algorithm C.2): listeners repeatedly re-sample labels (with replacement) from their perceptual encoding until they obtain M consecutive matching labels. This is the point at which the corresponding category label is determined and corresponding behavior executed. Since re-sampling is stochastic and the probability of drawing any particular label is proportional to its frequency in the encoding, average reaction times will depend on the initially encoded distribution of labels. If all labels in an encoding agree, then the representation contains implicit determinism: M matching labels will be re-sampled in exactly M steps (the fastest possible response). Whereas if labels are evenly balanced between alternatives in the encoding (the maximum uncertainty or maximum entropy distribution) then while it's still *possible* for a listener to re-sample M matching labels in exactly M draws, this is the state which will lead to, on average, the slowest responses. We note that this outcome is a stable property of discrete sampling, and it does not depend on any specific implementation choices or parameters. Because input near the category boundary is most likely to generate maximal-uncertainty perceptual encodings these, in turn, lead to the slow peak of the RT curve. The process of mapping from discrete perceptual encoding to behavior *must* lead to this output under DS and, as with the shape of the categorization curve, this does not depend on any particular value of N. In Figure 7 we show a comparison between RTs derived under DS with empirical data (Caplan et al., 2021; Kleinschmidt et al., 2015).

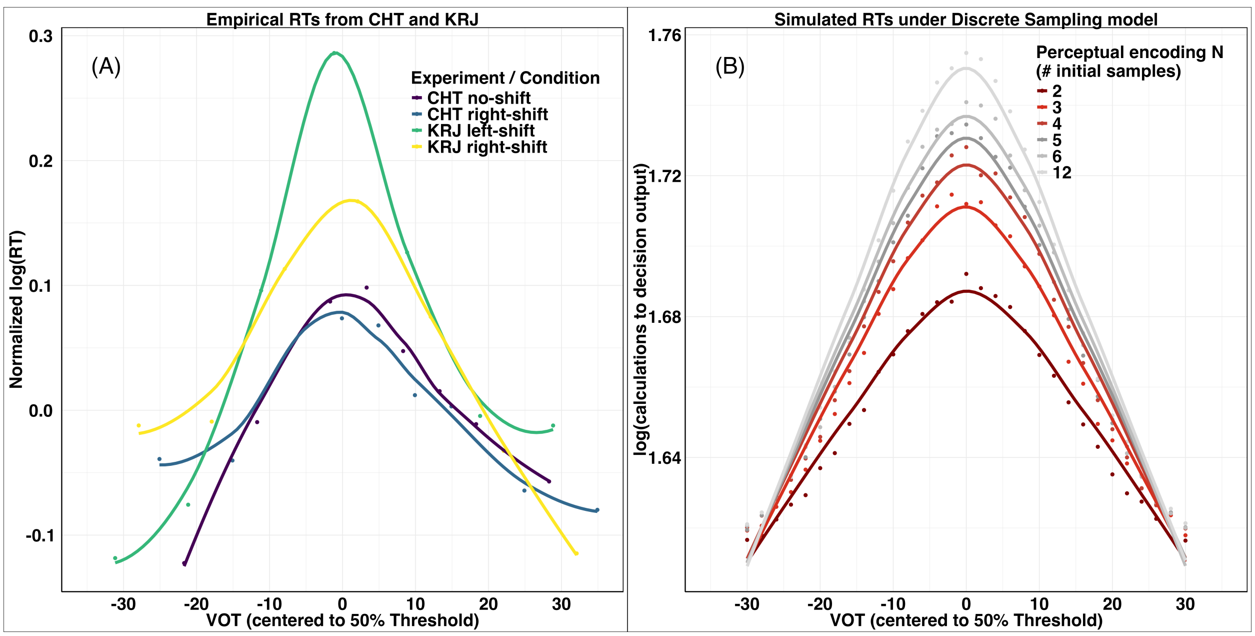

Figure 7: Left-panel (A): normalized (z-scored within each condition) log-transformed reaction time (RT) data from Kleinschmidt et al. (2015) and Caplan et al. (2021) as a function of phonetic input (VOT) centered on each category boundary; RT curve peaks neatly align to the category boundary differing only in peak magnitude. Right-panel (B): simulated log-transformed RT data from the DS model as a function of different encoding sizes N. The same concave RT curve, centered to the category boundary, is derived in each case even with the smallest possible values of N such as only two internal states (representing certainty or uncertainty)

This concave RT curve phenomenon appears in other tasks going back to Pisoni and Tash (1974); the auditory distance or confusability between two sounds has a corresponding effect on reaction time in making same/different judgments. For example, in an AX task, two sounds close to the categorical boundary will elicit longer RTs, while those further away from the categorical boundary will elicit shorter RTs. Interestingly, if the sounds are identical or if they are quite different, then the RTs are much faster. Thus we cannot make a general statement such as "shorter auditory distances lead to longer RTs" because this claim would result in judging *identical* stimuli to result in the longest RTs (which they do not). These facts are actually difficult to explain by a truly gradient representation sintce exact "identity" is no longer possible. In order to relax identity to extreme similarity or being "close enough" in the perceptual space to be called the same, then one would need to add an additional threshold in order to specify what counts as "close enough."

Overall, these observations suggest that the concave RT curve is not reducible to superficial aspects of the task or properties of the specific stimuli, but is instead a natural outcome of mapping from perceptual encodings to behavioral responses. Still, one might argue that the slowdown near category boundaries is a task artifact of another kind: if listeners internally encode speech in a continuous manner, then the RT peak may simply reflect difficulty of mapping a scalar representation onto a binary choice — e.g., the "distance" between a scalar and the effective "end points" in a 2AFC or AX task is greatest near the middle of the mental scale, captured by some drift-diffusion dynamic (Ratcliff & Rouder, 1998). On this view, RT differences should vanish under response paradigms that are themselves continuous, since there's no "discretization step" imposed by the nature of the task.

This contrast sets up a direct empirical test of divergent predictions. Under a continuous-encoding theory, reaction times in continuous response paradigms such as visual analog scale (VAS) judgments should be flat across the stimulus continuum: accessing a graded perceptual code should not depend on whether the intended response is near the midpoint or the endpoint of the scale. In contrast, DS predicts *the same concave RT slowdown* regardless of response format, since the locus of the effect is state of uncertainty in the discrete perceptual encoding, not the nature of the response itself. Recent data from Rizzi and Bidelman (2024) who examined VAS (and 2AFC) responses along a vowel continuum provide exactly the opportunity needed to evaluate these predictions and adjudicate between these alternatives.

4.3 Novel predictions of reaction-time patterns on VAS tasks

To evaluate the divergent predictions of gradient and discrete theories with respect to reaction time dynamics on continuous response tasks, we turn to data from Rizzi and Bidelman (2024). The authors directly compared binary (2AFC) and continuous (VAS) responses in a within-subjects experiment that crossed response type with a manipulation of signal-to-noise ratio. Stimuli consisted of five vowel tokens synthesized on a continuum between /u/ and /a/ varying linearly in F_1 (following Bidelman, Moreno, and Alain (2013)) and presented in a clean listening or high-noise context (signal-to-noise ratio [SNR] 100 vs. -2.5dB). Here we focus on reaction time dynamics under the VAS with further analysis of the relationship between paradigm, noise, and resulting categorization curves in Section 4.4.

On each trial, listeners were tasked with identifying the vowel present in auditory input using one of two paradigms, run in separate blocks: (i) a traditional two-alternative forced choice (2AFC) keypress task, and (ii) a visual analog scale (VAS) task where participants indicated their percept by selecting along a continuous bar labeled “oo” and “ah” at the endpoints. Crucially, both tasks used identical stimuli — differing only in response format — and all conditions were within-subject manipulations. Under a continuous encoding account, RTs in the VAS condition are predicted to be relatively flat across the continuum, since participants could directly map the “gradience” of their internal representation to a graded response. If perception were fully gradient, there would be no “decision threshold” which slows RTs at intermediate steps along the continuum. In contrast, the discrete sampling model predicts that the characteristic concave slowdown at the category boundary should emerge in both tasks, because the RT dynamics are driven by the uncertainty structure of the internal encoding, not by the external response format. Results are shown in Figure 8 and Figure 9.

Data from Rizzi and Bidelman (2024) provide striking support for the predictions of the DS theory, in contrast with a continuous encoding account. Across both 2AFC and VAS tasks, listeners’ RTs show the same inverted-U profile with the peak of slowest responses at the category boundary and faster responses towards the ends of the continuum. Importantly, this pattern is

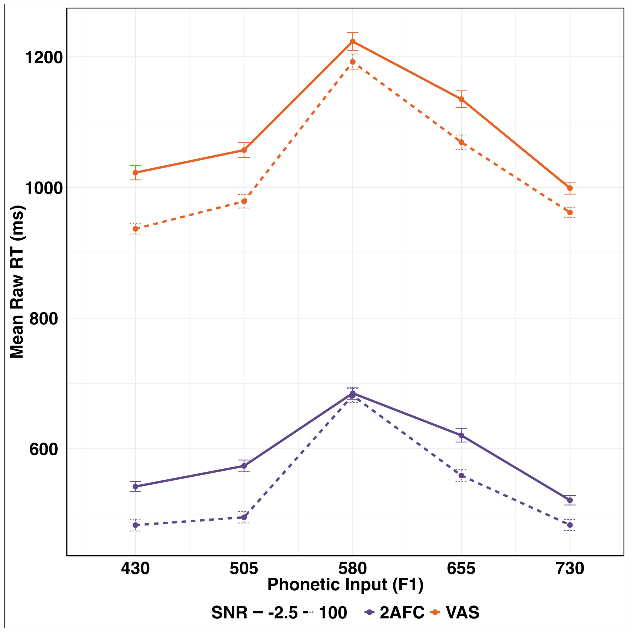

Figure 8: Reaction time data from Rizzi and Bidelman (2024). X-axis indicates equal sized step along the F_1 continuum between /u/ and /a/. Y-axis is raw reaction time in milliseconds. Color indicates response paradigm (VAS in orange and 2AFC in purple), while line-type shows the signal-to-noise ratio of the stimuli (solid lines are high-noise -2.5 SNR, while dashed lines are minimal noise 100 SNR)

present even in the VAS condition where participants had the opportunity to give a directly graded response. RTs were significantly slower at the continuum midpoint (mean of 1208ms) than at the two ends (mean of 980ms), $t(19) = 5.63, p < .001,$, 95% CI [143, 313], corresponding to a mean within-subject slowdown of about 23%. The presence of the boundary-peaked concave downward RT curve in VAS undermines the idea that the slowdown reflects a task-specific cost of discretizing continuous percepts into binary labels. Instead, it suggests that the slowdown originates in the structure of the perceptual encoding itself. The overall response time difference between the two tasks — with VAS responses taking about 500ms longer than binary responses — likely arises due to the speed-accuracy tradeoff in motor planning known as Fitts' law (Fitts, 1954; MacKenzie, 2018). It simply takes more time to plan and execute a longer mouse movement relative to a fixed key-press.

The concave reaction time (RT) curve, with maximal slowing near the perceptual category boundary, is only straightforwardly predicted by models in which speed is seen as a reflection of

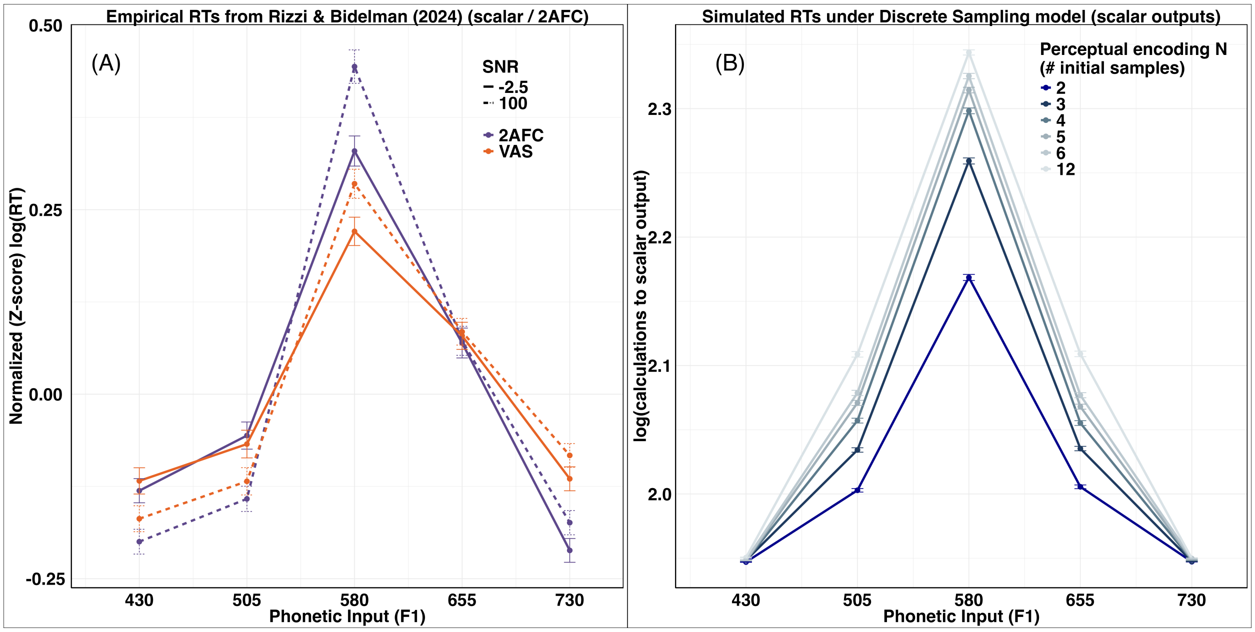

Figure 9: Left panel (A): Normalized (z-scored, log-transformed) reaction times from Rizzi and Bidelman (2024). RTs are slowest at the category boundary and faster toward the endpoints of the continuum. Right panel (B): Simulated log-RTs from the discrete sampling (DS) model generating scalar outputs across different perceptual encoding sizes N. The characteristic concave RT curve emerges robustly across all tested values of N, with the slowest responses corresponding to maximal uncertainty in the discrete encoding.

discrete sampling. On this account, stimuli near the boundary induce greater uncertainty, slowing the accumulation of evidence required for an output decision. By contrast, a gradient or continuous representation of speech does not inherently produce slower responses at intermediate points: if perception is genuinely continuous, RTs should be largely uniform across the continuum, and any apparent slowing in a forced-choice task could be explained as an artifact of mapping a graded percept onto discrete response options. Observing the characteristic concave RT curve even in a VAS task therefore provides compelling evidence that the slowing near the boundary is not a by-product of the task format or response modality. Instead, its persistence in continuous-response paradigms reflects an automatic consequence of perceptual processing near the point of maximal ambiguity, which is defined by the category boundary. This suggests that even when continuous responses are allowed, the perceptual system inherently represents speech in terms of discrete labels, and the concave RT curve is a fundamental signature of this category-based process. The concave RT curve here further manifests on participant responses to steps on a vowel continuum which some (see e.g. McMurray, 2022) have claimed lead to more "continuous processing." Nevertheless we see category-dependent behavior that is best explained via discrete uncertainty. With this understanding, we now turn to how task demands such as memory load, noise, and stimulus variability influence both categorization curve shape and RT dynamics.

4.4 Effects of paradigm demands and noise on categorization

As has been noted in the literature, e.g. Gerrits and Schouten (2004); Gur and McMurray (2025); McMurray (2022), the steepness of participants' categorization curves varies as a function of the experimental paradigm employed. Open-ended identification tasks (Ganong, 1980; Pisoni & Luce, 1987), or AX/ABX tasks (Liberman et al., 1957) tend to result in steeper response curves, while other paradigms such as the three-stimulus oddball (Carney, Widin, & Viemeister, 1977) — where the participant is presented with three stimuli and asked to identify the non-matching one — or 4IAX paradigm (Creelman & Macmillan, 1979) — where participants are presented with two pairs of sounds, one pair is identical and the other is different — lead to behavioral outcomes with notably shallower clines.

This appears to relate to the more memory-intensive nature of oddball/4AIX tasks compared with identification/AX paradigms. Indeed, analyses at the subject-level suggest that the strength of cortical responses in oddball tasks reflect individual differences in working memory capacity (Yurgil & Golob, 2013); and, in general, it seems that perceptual clines become less steep with a more memory intensive task (though the field's understanding of how complexity affects performance is still debated — see Liu and Li (2012) for a review).

One way to think about the differing effects of paradigm demands and noise on categorization is through variability or entropy. In an idealized world with no variability (the zero entropy situation), then every presentation of the same stimulus would result in the same outcome — a true step-function with the change in behavior occurring at the category boundary. The maximum variability (maximum entropy) scenario is one in which an input stimulus offers no predictive power over the resulting output — a uniform distribution over responses along the input continuum. This spectrum over entropy has a natural explanation via σ in the DS model. As σ, the variability over the applied category boundary per sample grows smaller ($\lim_{\sigma \to 0.0}$) this approaches the zero-entropy situation. Conversely, if σ were quite large (exceeding the range of the tested phonetic continuum), then the latent true category boundary would have vanishing effect on output decisions (*a la* the maximum entropy situation).

The DS framework thus affords a natural way to formalize, understand, and derive predictions from paradigm- and stimulus-modulated variability. In order to do so, we must consider the ways that perceptual variability may be realized. First, within-individual variability is the distribution over the differing outputs when measuring a single person on a single task (even the same stimulus) on multiple occasions. This sort of variability is well-attested across various timeframes, from short intervals such as seconds or minutes (Hultsch, Strauss, Hunter, & MacDonald, 2011) — precisely relevant for the current study — up through weeks and years. The latter is often in the context of aging, with increasing within-individual variation over time (Li, Huxhold, & Schmiedek, 2004). Across individuals, increased variability is strongly associated with decreased working memory and cognitive performance (Burnett et al., 2025; Hultsch et al., 2011). Second, when comparing variability across tasks we see the same trend. As task difficulty increases (due to memory constraints, or perceptual noise, etc.) so does variability in responses and the spread of the RT distribution, alongside a corresponding drop in accuracy (Donkin & Van Maanen, 2014; R. D. Luce, 1991; Overbosch, De Wijk, De Jonge, & Köster, 1989; Piéron, 1952). It has been proposed that variability results from transient lapses of attention or fluctuations in executive control (MacDonald, Li, & Bäckman, 2009), resulting in slower reaction times and lapse errors such as target detection failures. This effect is persistent: individuals who perform better on general cognitive tests tend to have more consistent scores across repeat tests (Campbell et al., 2025).

Under the DS model, we can actually understand the effect of memory and task noise on the perceptual function. Task complexity (or the load imposed on working memory) can be cashed out as more variability in the output due to limited resources (Gold & Shadlen, 2007; MacDonald et al., 2009). As memory load increases, noisy-input based decisions are likely to be less accurate, as memory resources are concurrently being used for other purposes. In mathematical terms, this means that the variance (σ) associated with the decision boundary effectively increases. Thus high-memory load contexts result in shallower clines than low-memory contexts even though the underlying DS model is the same in all cases, and always composed of wholly discrete internal states. This is shown under DS in Figure 10A; say that the variance related to the categorical boundary increases by 5 or 10 units in a high-memory load paradigm compared to a low-memory load paradigm, then the perceptual function is going to be less steep for the former.

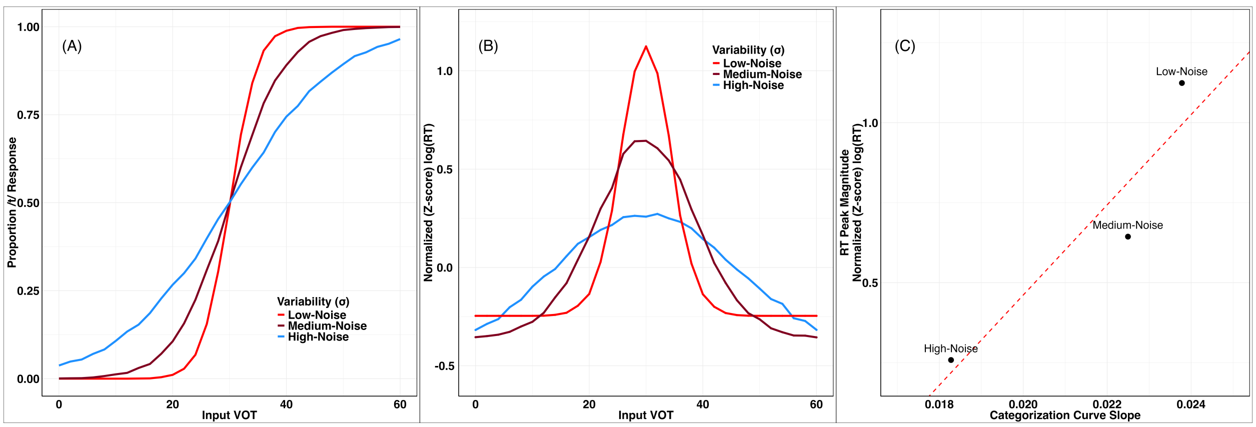

Figure 10: Categorization (left — A) and RT (middle — B) curves as a function of memory load (σ) in the DS model. The right panel (C) shows the systematic relationship between categorization slope and RT peak magnitude as a function of σ; increased noise/variability in the application of the category boundary to each sample naturally leads to both flatter categorization curves and correspondingly shallower RT peaks near the point of maximal uncertainty.

In addition to paradigm difficulty (e.g. the increased memory demands of 4IAX relative to a simple AX task), variability may increase due to stimulus-level facts such as literal noise. Assuming the sources of such complexity are independent, the effect of the additional sources of variability should be additive. Consequently, the variance related to the categorical boundary will be higher. As noted (Section 4.3), Rizzi and Bidelman (2024) manipulated both response modality (2AFC vs. VAS) alongside the stimuli signal-to-noise ratio (SNR 100 vs. -2.5). In other perceptual domains, such as vision, it is well-known that signal-to-noise ratio is positively correlated with accuracy, and correspondingly, negatively correlated with both response times and response variability (Palmer, Huk, & Shadlen, 2005). The key point for this present inquiry is that multiple manipulations (task-load and SNR) will have an additive effect on response variability, and that (σ) has a joint impact on both output (categorization or VAS responses) and the RT distribution via a single, shared pathway. See Figure 10: the DS model predicts that easy, low-noise conditions will lead to a sharper categorization curve in tandem with a larger relative RT slow-down at the category boundary[13]. Panel C shows the predicted systematic relationship between the two output variables.

With this in mind we turn again to data from Rizzi and Bidelman (2024). The authors exposed participants to a vowel continuum using a within-subjects manipulation of binary (2AFC) or continuous (VAS) response crossed with high-noise (-2.5dB SNR) or low-noise (100dB SNR). Beyond simply the presence of a concave RT curve or a sigmoidal categorization curve, the DS model makes a critical, quite specific prediction which relates the inter-sample variance (σ), as affected independently by response type and SNR, to both processing speed and response accuracy. DS predicts that both the relative magnitude of the RT peak *and* the perceptual sensitivity (categorization curve slope) should co-vary in tandem across the four experimental conditions. In this framework, each of these two output measures (RT peak magnitude and categorization slope) should exhibit the exact same ordinal pattern (Equation (3), where $\succ$ indicates both a higher relative peak in RT magnitude and also a greater categorization slope)[14]:

$$\text{2AFC low noise} \succ \text{2AFC high noise} \succ \text{VAS low noise} \succ \text{VAS high noise} \tag{3}$$

We see this prediction borne out precisely. Figure 11A shows the expected four-way ordering of categorization slopes as a function of response type and SNR: conditions which presumably

[13] "Relative" here refers to the normalized peak; that is the magnitude of the difference between the slow RT peak and the stimuli generating the fastest responses. Normalization shows this *relative* difference rather than the raw RT at the slowest point

[14] The exact ordering in 3, rather than the transpose where 2AFC and VAS get interleaved, depends on the relative magnitude of the noise induced by the two effects. If we assume that response modality has an outsized effect relative to SNR then we predict this precise ranking

increase σ flatten the curve. Concurrently the same conditions have a corresponding effect on RT relative peak magnitudes, with the sharpest-curve condition generating the largest RT peak, fully matched down to the flattest-curve condition having the smallest relative RT peak (Figure 11B). Note that there are 24 possible ways that four conditions like this could have been ordered, yet we see the exact same ranking of curve-slopes in alignment with RT-peak magnitudes as predicted by DS. Figure 11C shows the direct correlation between the two output measures, where a simple simple linear regression between categorization slope and RT peak magnitude accounts for 83% of the variance across the four conditions. We further note that while the exact slope of the categorization curve or the relative magnitude of the RT peak individually could be fit by adjusting σ, within DS they *cannot* vary independently; the model can *only* derive exactly the ordering we see in the empirical data (assuming SNR and VAS both add variance to sampled encodings). If these two output measures could vary independently of one another it would be vanishingly unlikely to see exactly this empirical outcome ($\sim P(0.0017)$ based on 24 possible permutations of four variables and generating the same outcome twice to produce this exact alignment.) Noise of all sorts plays a large role in measurable curve shape, and we must exercise caution in inferring gradient representations on the basis of the slope of the response function — a point we return to regarding the use of neural data in speech perception (e.g. Getz & Toscano, 2021; Toscano et al., 2010) in the General Discussion.

Discrete sampling provides a simple explanation of variability across trials, tasks, and stimuli. This variability does not reflect “different strategies” for some paradigms or particular phonological contrasts nudging listeners to “behave gradiently” or “behave categorically.” Our view rejects the idea that “listeners perceive speech more categorically in some tasks than in others” (Kong & Edwards, 2016, p. 40) if the object of study is a psychological process rather than the behavior which it generates. Instead, variability reflects in-moment difficulty and variable resource use operating over a single core mechanism (discrete sampling) which underpins the perception of speech. Next, we move from task-based and within-subject variability to between-subject variability. We further demonstrate that the apparent distribution over participant responses is also consistent with

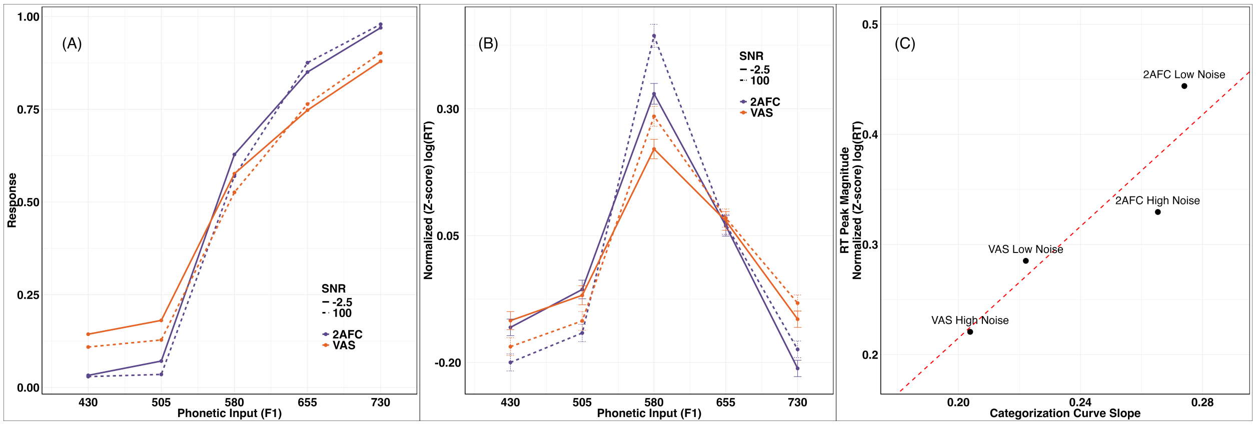

Figure 11: Data from Rizzi and Bidelman (2024). Categorization (left — A) and RT (middle — B) curves as a function of response-type / SNR condition. The right panel (C) shows the systematic relationship between categorization slope and relativeRT peak magnitude as a function of condition.

and explained by DS.

4.5 Predicted distribution of apparent inter-subject variability

In recent years, the issue of individual differences has become important both in the realm of speech (Fuchs, Pape, Petrone, & Perrier, 2015), and other perceptual domains such as vision (Mollon, Bosten, Peterzell, & Webster, 2017), touch (Campagna & Chamberlain, 2024), or pain sensation (Nielsen, Staud, & Price, 2009). Narrowing our focus to speech perception research, and in particular focusing on how participants perceive phonetic continua, participants have been observed to have differing slopes for their categorization functions, with some participants exhibiting steeper slopes, and others with shallower slopes (Hazan & Rosen, 1991; Kapnoula & McMurray, 2021; Kong & Edwards, 2016; Kutlu, Baxelbaum, Sorensen, Oleson, & McMurray, 2024; Ou & Yu, 2022). Such results have been interpreted as evidence that speakers are “more” or “less” categorical in how they perceive the input (Kapnoula & McMurray, 2021; Kong & Edwards, 2016).

However, there are at least two possible ways in which one can understand such individual differences in the behavioral output. The first is to indeed infer that the observed differences are from systematic quantitative differences in cognitive systems of the listeners. However, we find claims about quantitative differences in categorization functions an unusual emphasis if our goal is in building fundamental theories of psychological processing.[15] Alternatively, the observed individual differences in behavioral output may reflect random variation, as would be the case for any sampling distribution of a random variable. If so, the observed variance does not inform us about any substantial differences in the cognitive systems of the speakers.

One can also lay out distinct empirical manifestations between the two possibilities above, based on the excellent discussion in the recent statistics literature on the topic of individual variation (Senn, 2014). In short, in order to establish the first possibility of actual quantitative differences in the cognitive systems of the listeners, one needs to conduct *independent* replications of the experiment on the particular listeners and show that they consistently show the same differences, despite controlling for participant differences in other factors that interact with the cognitive system under study. We believe this has rarely been done in the relevant literature (though see Basu Mallick, F. Magnotti, and S. Beauchamp (2015) for a compelling example of such stable, individual variation). Furthermore, in the absence of such evidence, the simpler (null) hypothesis is to maintain that the observed variation is due to random sampling variation. This, of course, is not to deny that people *can* vary. But we must evaluate whether the variation we see differs from what would be expected simply due to sampling a random variable. DS provides a concrete model from which to derive trial-level (and subject-level) outcomes, thus allowing us to see whether the attested distribution of inter-subject variation on some experiment meaningfully diverges from a reasonable null hypothesis expectation.

Here we compare the expected distribution under DS to empirical data from the /t/-/d/ categorization experiment in Caplan et al. (2021). We fit separate logistic regressions for each subject to predict participants' trial-level responses based on VOT. This was done both for the human empirical data as well as simulated participants in the DS model. To maximize comparability, the DS simulation exactly matched the steps along the VOT continuum, the number of responses

[15]To us, the issue of individual variation should focus on *qualitative* differences rather than merely quantitative ones, which might naturally result from natural sampling variation, if they share a single underlying cognitive system. For example, there is also, for instance, substantial variation in the heights of different people across the world — including systematic differences depending on where in the world one looks — but no biologist would describe children as undertaking different "growing strategies."

per VOT-level, as well as the number of total subjects from Caplan et al.'s design. We further matched the empirical category boundary μ with the only free parameter thus being the boundary standard deviation σ. Using "Earth Mover's" distance (Wasserstein distance; Villani (2008)) as an objective function to select σ best approximating the empirical data we calculated the slope of each response function (i.e. one slope per subject) and ranked them all from steepest to shallowest clines; separately in the empirical data and DS-derived data.

Results are shown in Figure 12. Panel A shows the full distribution over subject-level categorization slopes, with the flattest subjects (5^{th} percentile and below) indicated in blue and the sharpest subjects (95^{th} percentile and above) indicated in red. To assess whether meaningful qualitative differences between participant slopes (distinct sub-groups) was present, we conducted a Hartigan's Dip test and found no statistical evidence to support non-unimodality of slopes ($D = 0.0168, p = 0.9189$)

The spread of the distribution over participant behavior is predicted extremely well by DS. The Wasserstein distance between the empirical distribution and that generated under DS is ≈ 0.018, meaning that amount of slope-shift required to exactly convert the DS-derived distribution into the empirical distribution from Caplan et al. is less than 0.02 on average. In Figure 12B-C we zoom in to the flattest and sharpest subject groups (bottom and top 5% of subjects respectively): transparent lines show individual subjects while the opaque lines show the group average — solid line is empirical subjects from Caplan et al. (2021) while the dashed lines are the corresponding DS groups. Across the board, the DS model derives exactly the variation we see with respect to categorization response. Not only is the limited number of empirical subjects' data showing a comparatively flat response curve not evidence against discrete representations, this is actually predicted to occur at exactly this rate under the DS model.

To provide some intuition behind this result, this must occur because the number of trials per participant is finite. Since each VOT-level was tested a limited number of times, even if every trial were independent we should still expect to see a degree of systematic-looking variation measured at the subject-level. Just by accident some participants will end up producing responses that appear

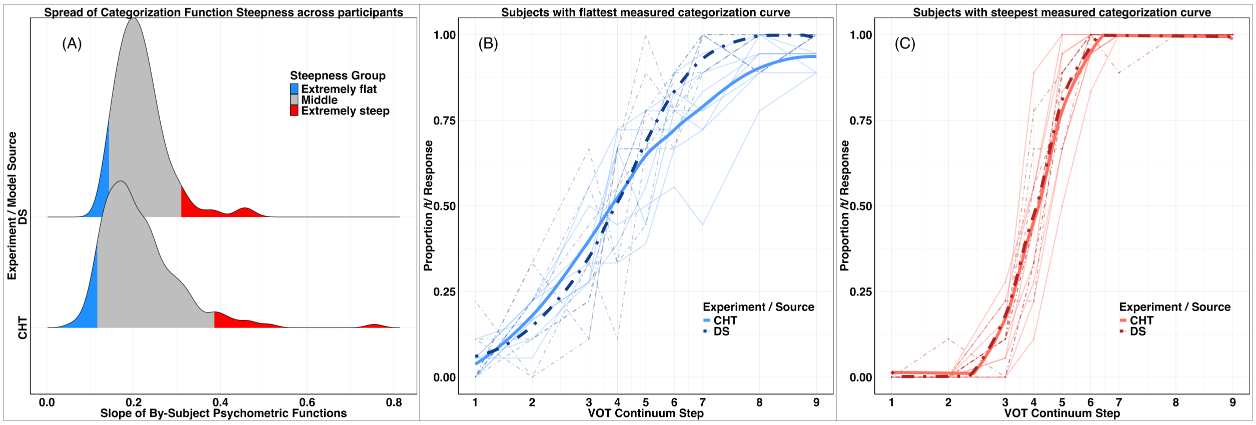

Figure 12: Left (A) shows the ridge density plot of the distribution over subject-level categorization slopes in Caplan et al. (2021) (CHT) and under DS. Blue colored region indicates the lowest ventile, while the red region shows the highest ventile in terms of categorization function slope. The middle panel (B) zooms in to the blue-region showing a comparison between the flattest 5% of participants in CHT overlaid with the flattest 5% simulated participants under DS. The right panel (C) shows the same thing but for the 5% steepest-curve participants in CHT and DS.

flatter or sharper despite operating with a shared cognitive architecture.

To be clear, we are not arguing that this category boundary is innate, nor that all subjects must share exactly the same boundary; indeed, we know from the large literature on perceptual learning that category boundaries are malleable up to a point. Rather, whatever the learned differences across participants may be, they appear to be negligible at least in the set of experiments under analysis here. We also do not aim to deny that listeners must, in fact, vary to some degree over their available cognitive resources; this appears to be the primary driver in between-subject differences in response variability (Campbell et al., 2025; Li et al., 2004). Nevertheless, we do argue that these inter-subject differences are quantitative (via higher or lower σ) rather than qualitative (a subset of participants representing speech in some way other than discrete states). The results thus far are consistent with all listeners sharing a unified architecture for the discrete perception of continuous speech subject to noise in the category boundary applied; and the DS model does a remarkably good job at predicting overall outcomes even absent any curve fitting to estimate σ beyond the group-level.

4.6 Summary of results

Across the multiple analyses presented above, a consistent pattern emerges: discrete sampling provides a unifying account of speech perception that captures both aggregate and trial-level behavior. Category judgments — whether binary or continuous — exhibit sigmoid curves whose shape is independent of the size of the perceptual encoding (N), demonstrating that discrete representations are sufficient to produce the gradient output trends observed in both 2AFC and visual-analog scale (VAS) tasks. At the same time, the shape of these curves alone is equivocal with respect to the nature of the underlying representation; it is only by considering reaction-time dynamics that the discrete nature of perceptual encoding becomes apparent.

The hallmark concave downward RT curve, dating back to Pisoni and Tash (1974), is robust across experimental paradigms, phonetic continua, and response modalities. Critically, RTs are slowest precisely at the category boundary, an effect that shifts accordingly as a result of perceptual learning. DS naturally, and uniquely, predicts this relationship as a function of uncertainty in the perceptual encoding across both binary and continuous response tasks. This phenomenon is not an artifact of task demands or from mapping a perceptual encoding to a categorical output measure, but rather a fundamental signature of the underlying discrete encoding process. As with categorization curve shape, this concave relationship holds regardless of N or M (the threshold for evidence accumulation). Moreover, experimental manipulations of SNR and response modality independently and systematically affect both categorization slope and relative RT peak. This is in precise accordance with DS predictions: higher noise or task difficulty flattens the categorization curve while simultaneously attenuating the relative RT peak, while lower noise preserves steep curves and more pronounced RT slowdowns.

Finally, apparent differences across participants — some "sharper," some "flatter" — is predicted by DS even with no curve-fitting. The number of subjects, trials, and category boundary are all properties determined by the experiment. The attested distribution of observed categorization slopes across participants, including the extremes, emerges naturally from a shared architecture for discrete perception operating under trial-level stochasticity in the application of the category boundary.

In sum, the converging evidence from categorization behavior, processing speed, task demands, and inter-subject variability support a single, parsimonious idea: speech is perceived via discrete states. Listeners map continuous input into category-based internal representations via a simple sampling procedure. With this foundation, we now turn to a discussion of the broader implications for our understanding of speech perception and cognitive architecture (Section 5).

5 General Discussion

Across the empirical sciences, core progress has often come not from more precise measurement of the same behavioral patterns, but by identifying new dimensions of evidence that distinguish among otherwise observationally equivalent theories. The present work applies this principle to one of the central questions in the psychology of language: how perceptual systems encode continuous sensory input into mental representations supporting symbolic inference. We have argued that the traditional opposition between "categorical" and "continuous" perception lumps together two independent dimensions — the *content* of representations (cue-based vs. category-based) and their *granularity* (binary, discrete (4 ± 2), or continuous (arbitrary precision)). By making this distinction explicit, and by formalizing a discrete yet uncertainty-preserving architecture, the Discrete Sampling (DS) framework shows how graded behavioral data need not arise from gradient mental representations. Rather, a system — instantiating a discrete, category-based theory as in Figure 2 — composed entirely of discrete internal states constructed by sequentially sampling the input can uniquely generate (and predicts) the categorization curves, reaction time dynamics, and the task- and stimulus-dependent behavior that are characteristic of speech perception.

DS thus reframes the debate over the perceptual representation of speech as one of *algorithmic sufficiency* rather than *phenomenological appearance*. What matters is not whether behavior appears graded, but how the underlying architecture supports memory, uncertainty, and downstream inference once the input signal has vanished. In this sense, the DS framework parallels the distinction between weak and strong generative capacity (Chomsky, 1963): it is not enough for an account to reproduce the surface distribution of responses; it must also specify the algorithmic structure that makes such behavior possible. This aligns nicely with the computational vs. algorithmic distinction in Marr's levels of analysis (Marr, 1982). Yet it is a computation-level constraint — inherent to the physical systems involved — that perceptual encoding serves as a bottleneck through which all subsequent cognitive operations must pass (Caplan et al., 2021; Christiansen & Chater, 2016): only information preserved in this intermediate representation can be integrated, learned from, or used for decision making. The Discrete Sampling model is consistent with the "now-or-never bottleneck" which Christiansen and Chater (2016) identify, and achieves the promise of integrating continuous input with discrete representations highlighted in that work. This not only addresses a long-standing empirical question in speech research but also clarifies a broader principle for cognition: that continuity in behavior need not entail continuity in representation.

A general theme may be found here in genetic variation. When phenotypic traits are measured within a population, the data often form a continuous distribution (Darwin, 1859); the length of pigeon beaks, say. Yet the biological basis of these gradients is unambiguously discrete: genes composed of discrete nucleotide (DNA) sequences. The apparent continuity of phenotypes reflects the joint influence of many individually discrete genetic factors interacting with environmental variability, not a fundamentally continuous code. As it is in the mind: infinite variation and continuous gradation may be composed of wholly discrete parts. The gradience of speech perception behavior stems from discrete representational units operating over noisy and variable sensory input.

5.1 Fuzzy behavior and uncertainty

This view, in fact, fits nicely with the large literature describing gradient ***behavior*** in speech perception. Within-category differences in VOT have, when averaged, a gradient effect on fixations in the visual-world paradigm (McMurray et al., 2002). As described by the authors "fine-grained acoustic/phonetic differences are preserved in patterns of lexical activation." This is true if taken very precisely. Yet it is a small, tempting leap to fill in the gaps here and infer that the mechanism by which phonetic gradation survives via group-level referent-fixations is a continuous perceptual encoding. This logical jump is not warranted, and as we have demonstrated the discrete perceptual states probabilistically generated under DS would derive exactly the attested visual-world paradigm effects.

Similar trends can be seen in continuous action paradigms. Spivey, Grosjean, and Knoblich (2005) tracked participants' mouse movements while they responded to spoken words in the presence of phonological competitors. Even when participants ultimately selected the correct referent, their hand trajectories initially curved toward competitor items, revealing a continuous pull that reflects early activation of multiple lexical candidates. Under DS, early labels sampled from input are consistent with all cohort-items on the screen with behavior which reflects that. Samples generated from later-arriving input disambiguate, with accumulated evidence culminating in the correct selection.

There is a through line here which we believe can be traced back to the early evidence for graded behavior in speech perception: Massaro and Cohen (1983b)'s perception-as-fuzzy-logic approach. Massaro and Cohen propose that listeners compute partial activation across multiple categories, effectively representing uncertainty in a continuous fashion. Notice that this is *nearly* identical to the framing we provide here, modulo the word "continuous." On our view, the description of activation/uncertainty as continuous rather than discrete is really epiphenomenal: the observed gradience arises not from inherently continuous perceptual states but from a distribution over discrete category samples. Specifically, if each trial involves a probabilistic selection of discrete labels with a non-zero variance ($\sigma > 0$), the aggregate behavior reproduces the smooth categorization functions described by Massaro and Cohen (1983b). DS provides a concrete mechanism by which the original spirit behind fuzzy logic — representing uncertainty in perception — may be cashed out.

Another clear line of evidence for maintenance of uncertainty in speech perception comes from long-distance "cue-integration." When input is locally ambiguous, listeners are able to integrate information from temporally separated cues before committing to a category (Bushong, 2025; Connine et al., 1991). Much of the existing literature, however, conflates gradient behavior with gradient representations. Authors sometimes acknowledge that uncertainty could be stored in various potential forms while simultaneously endorsing a continuous-cue interpretation, e.g., "These results suggests that listeners are able to maintain relatively fine-grained details about prior linguistic input over long perceptual timescales." (Bushong, 2025). We know this need not be so. Graded behavioral outcomes emerge naturally from discrete, category-based perceptual states that encode uncertainty. Previous studies often use the terms "subphonemic" or "subcategorical" as shorthand for the behavioral signature of within-category sensitivity. Under DS, these perceptual encodings represent information about or between categories, making "supracategorical" a more precise term for the kind of sensitivity observed.

The key insight here is that we must distinguish an input-output mapping from the psychological process which generates behavior from sense data. Where during the perceptual process categories come online is important since it is these discrete symbols which are the elements of inferences. "[i]f linguistics can be said to be any one thing it is the study of categories: that is, the study of how language translates meaning into sound through the categorization of reality into discrete units and sets of units" (Labov, 1973, p. 342). What's more, the acquisition of phonological categories appears to be primarily governed by discrete/symbolic distinctions at the lexical level rather than continuous acoustics (Cui, 2020).

5.2 Insights from ERP response patterns

Another line of evidence that has been used to argue for continuous perceptual representations comes from electrophysiological work looking at a specific ERP component, namely, the N100/N1 or P3/P300 (Getz & Toscano, 2021; McMurray, 2022; Toscano et al., 2010). To take a representative example, Toscano et al. (2010) presented listeners with two synthesized VOT continua: "$\{/b/-/p/\}$each" and "$\{/d/-/t/\}$art", where the VOT was increased in 5ms steps from 0-40ms. In a standard mismatch paradigm, participants had to press a button for the target and non-target stimuli. Participants' ERPs were recorded from onset of the stimulus, and multiple electrode sites were employed to get an average ERP response. The averaged N1 response to VOT changes varied linearly as a function of VOT, while participants' behavioral responses were sigmoidal. This observation was taken (as are ones like it) as evidence that categorization reflects internal discreteness but perception, at its source, depends on a perceptual encoding that is continuous. How should we make sense of this finding in light of the present discussion?

The interpretation generally provided for such findings (Getz & Toscano, 2021; Toscano et al., 2010) is to view perception (and encoding) as a two-stage process (as illustrated in Figure B2). On this view, there is an initial, short-lived stage (up to, say, 200 ms post-stimulus) during which the input is represented continuously (here linearly) before subsequent transformation into its more stable, discrete counterpart. Such a two-stage view accounts for the early linearity of the N1 response, though a later discrete stage is still required to feed into a behavioral or categorization decision. The reaction time dynamics cannot easily be derived without such an eventual discrete encoding. In such a multi-stage view though, the timing of discrete-category information must also be available extremely early; Sharma and Dorman (1999), for instance, found sensitivity to voicing categories in the N1 response — i.e., observable in the first 100 ms after stimulus presentation. A range of work using the mismatch negativity paradigm highlights the presence of (discrete) category structure in ERPs in a similarly early 100-300ms window post stimulus (e.g., Hestvik & Durvasula, 2016; Kazanina, Phillips, & Idsardi, 2006).

We suggest that a more parsimonious explanation is possible — one that is fully compatible with the aforementioned electrophysiological data while avoiding commitments to multiple representational "stages" or an additional function to translate between them. That is, these ERP patterns are exactly what we should *expect* to occur under Discrete Sampling. Recall that the physical and behavioral constraints motivating a limited granularity of encoding (via JNDs, subitizing, and cortical temporal resolution) apply equally here. ERP waveforms reflect averaged electrophysiological activity across many trials; they do not provide a direct, trial-level readout of an individual perceptual encoding. As we have emphasized throughout this paper, and in line with e.g., Estes

(1956); Sidman (1952), one cannot distinguish gradience from discreteness on the basis of curve shape. Averaging over stochastic discrete states naturally produces smooth trends, including those modeled as linear, even when the underlying representations are categorical.

ERP responses are comparatively noisy, owing to biological variability, aggregation over large neural population, and the physical noise associated with scalp measurements. In DS, large increases in noise naturally lead to linear response functions as with increasing σ (recall Section 4.4). A linear relationship between, for instance, VOT and ERP amplitude, does not require a continuous internal perceptual code — it arises naturally whenever discrete internal states are generated through a high-noise physiological signal. The account here does not deny that the auditory system processes continuous acoustic input or that measured neural responses vary in graded ways. Rather, DS situates these facts within a framework in which the representational building blocks for perception encoding and categorization remain discrete.

Ultimately, any theory of speech perception must be compatible with the biological mechanisms that translate continuous acoustic input into cognitive representations. We believe that evidence strongly supports a view that this transformation is realized through discrete mechanisms: at the bottom-level of implementation individual neurons operate in an all-or-none manner. A cell either generates an action potential or it does not. The apparent continuity seen in neural recordings arises not from partial activation within neurons, but from the probabilistic, population-level activity of many discrete units an in the lightbulb-problem (Gallistel, 2021). Graded neural signals reflect the dynamics of large ensembles of binary components (Rolls & Treves, 2011). A speculative analogy may be made that the Discrete Sampling framework parallels the organization of its neural substrate: both exhibit continuous outcomes that are, at their core, the product of discrete, stochastic events.

5.3 A broad psychological principle

It is an important open question how broadly generalized the discrete nature of perceptual encoding is. The specific inquiry thus far has been within the confines of speech perception, though we see no *a priori* reason why this framework should be limited to any particular modality. Similar issues arise in vision: for example, observers must transform low-level cues of hue or brightness into perceptual encodings that can support higher-level category judgments (Lennie & D'Zmura, 1988). Behavioral work on color perception likewise shows graded mappings from physical stimuli to category responses (Bornstein & Korda, 1984; Huette & McMurray, 2010), raising the possibility that an analogous discrete-sampling mechanism might underlie visual categorization. More broadly, this question invites investigation into other perceptual and cognitive domains — e.g., including shape, emotion, or motor control — and the mechanisms by which discrete, category-based representations are supported across modalities (Angeli, Davidoff, & Valentine, 2008; Fugate, 2013).

Irrespective of the precise cross-domain generalizability, the implications of discrete perceptual encoding extend beyond speech. Because encoding occurs online and under strict temporal constraints (Caplan, 2021), the resulting intermediate representation functions as an information bottleneck: once the raw input is gone, all subsequent inference, learning, and decision-making must operate over this representation. The content and format of these representations therefore determine which aspects of sensory input remain accessible to later processes. Although much of the present discussion has focused on speech, the same logic extends to perception in other modalities: continuous visual input, for instance, must be transformed into discrete, structured representations before it is able to support object recognition, motion inference, or other higher-level computations. The Discrete Sampling framework thus offers a domain-general account of how perceptual systems transform continuous input signals into symbolic, combinatorial representations suitable for cognition.

Beyond humans, (continuous) probabilistic models of associative learning have been employed, with great success, to account for animals' responses to environmental uncertainty (e.g., Bush & Mosteller, 1953; Wagner & Rescorla, 1972). While such models offer strong predictive power to behavior, they provide little constraint on the underlying cognitive mechanisms: specifically, ***how*** are the posited probabilities represented within an animal's cognitive system and what algorithmic process supports updating and comparing such values? It is possible that the Discrete Sampling framework provides insight here. A mental encoding with N labels provides a discrete approximation of probabilities; a potentially better approximation as N increases. Resulting animal actions are governed by the chance of drawing one sample from the representation or another. The "true" probabilities are derivative, arising from internal discreteness. Future work could explore how different procedures for adding or removing labels from the representation yield distinct probabilistic learning outcomes, or how behavioral differences on reinforcement learning tasks relate to the appropriate choice of N for a given capacity.

A complementary perspective on this question concerns the role of the evidence accumulation parameter M within the DS framework. While biologically constrained, decisions about how much evidence to sample before committing to a decision may not be purely mechanical, but also tied to relevant value considerations (Gold & Shadlen, 2002, 2007). If speed is prioritized (or required in some context) perhaps fewer samples are re-drawn (a smaller M); if accuracy is paramount then more sampling is warranted. This picture is consistent with a view in which, even in perceptual decision-making, the exact shape of behavioral curves will depend on real-time goals and physical constraints.

At the same time, it is reassuring that the qualitative behavior of the DS model is remarkably robust to the precise values of N, σ, or M. We see this robustness or "inflexibility" as a theoretical strength: DS does not rely on specific parameter-tuning to derive the attested empirical phenomena, but instead generates its predictions from a tightly constrained underlying architecture. Nevertheless these parameters are best understood as empirical reflections of properties of the external environment and of individual cognition rather than as free knobs to be adjusted post hoc. On a forward-looking note, the DS framework affords the possibility for studying how separately fitting N or other parameters to particular tasks or participants may offer a window into the locus of latent individual differences.

Stepping back from particular parameter values, we find a deeper theoretical issue. Discrete perceptual representations can be thought of as strict subsets of gradient perceptual representations. Given that they form strict subsets, and thus form smaller hypothesis spaces, discrete representations are more constrained by definition. This asymmetry has important implications for theory construction. A more flexible, gradient system (the *superset* model) will nearly always be able to be curve-fit to a larger range of data. But this flexibility is not a virtue in itself and the constrained nature of discrete representations is an asset rather than a liability: it forces the model to make stronger commitments, providing more informative explanations when those predictions succeed. This line of reasoning aligns with longstanding arguments in the philosophy of science that simplicity and parsimony are virtues of theory (Goodman, 1943), and it underscores why the empirical success of DS relative to a more flexible, continuous alternative is theoretically important.

Returning to the discrete maintenance of perceptual uncertainty, a closely related idea applies to early human word learning. Computational approaches such as Pursuit (J. S. Stevens, Gleitman, Trueswell, & Yang, 2017), or contemporary variants (Yue, LaTourrette, Yang, & Trueswell, 2023), highlight the need to maintain uncertainty about potential referents or hypothesized word meanings over time. Crucially, however, this uncertainty is not instantiated as a true, continuous probability distribution, but emerges from discrete, integer-based evidence accumulated for specific candidate hypotheses. In this sense the learner, averaged across learning trials, often behaves *as if* engaging in probabilistic inference while calculating over fundamentally discrete internal representations. This explanation, and DS in general, has a natural symbiosis with the Rank hypothesis for lexical access (Murray & Forster, 2004): (log) frequency is a continuous measure which lexical process speed appears sensitive to, yet its cause may nevertheless be rooted in a discrete representational format. The perspective explored here thus unifies otherwise disparate phenomena: the same principles that govern perception may also underpin learning.

A critical challenge for future work will be to understand *how* perceptual systems implement learning and adaptation within the constraints of a discrete representational architecture. Perceptual learning effects are robust and easily induced (A. G. Samuel & Kraljic, 2009), yet the underlying mechanisms that actually support them remain poorly understood. Under DS, such effects may arise through shifts in category boundaries, whereas σ reflects noise in perceptual application rather than a learned parameter that can be strategically exploited (as in Bayesian accounts).

These separate pathways may lead to differentiating predictions for DS as a mechanism for perceptual learning relative to general distributional learning (Clayards et al., 2008; Kleinschmidt & Jaeger, 2015). Similarly, short-term adaptation effects like those induced under a lexically-guided perceptual learning paradigm are known to fade, even during a brief test phase in the absence of feedback. Do such phenomena share a common underlying mechanism, or are they merely superficially similar? How does this relate to classic selective adaptation effects (Eimas & Corbit, 1973)? These questions may also intersect with the Perceptual Magnet effect (Grieser & Kuhl, 1989; Iverson & Kuhl, 1995; Kuhl, 1991; Kuhl, Williams, Lacerda, Stevens, & Lindblom, 1992), which may itself reflect properties of the categorization process rather than specialized warping of perceptual space (Feldman et al., 2009; Iverson & Kuhl, 2000; Lotto, Kluender, & Holt, 1998). Exploring these issues represents a promising direction for future research, one that could clarify the algorithmic principles governing the formation, maintenance, and flexibility of discrete perceptual representations across speech, vision, and other domains.

To that end, we see Discrete Sampling not as a standalone answer to a specific question but as a potentially broad research program — a framework for thinking about how continuous behavior emerges from discrete cognitive states across tasks, domains, and species. If continuity in behavior is the surface expression of discrete internal states, then we would like to emphasize a commitment to separating the prediction of measurable behavior and an explanation of its internal cause. Isn't the ultimate goal for cognitive science not merely to describe what the mind appears to do, but to understand ***how*** it does so?

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A Categorical Perception

The original goal of Liberman et al. (1957) to examine *"whether or not, with similar acoustic differences, a listener can better discriminate between sounds that lie on opposite sides of a phoneme boundary than he can between sounds that fall within the same phoneme category"* — simply if there is an effect of category boundary on discrimination. We find it puzzling that this has been so misrepresented in the subsequent literature; what is typically taken to be the extreme-CP argument was originally used in used Liberman et al. (1957) only as a quantitative evaluation as a null model. As stated: *"In addition, an attempt has been made to determine to what extent the relation between discrimination and labeling has here been reduced to its theoretical limit. For that purpose, the obtained discriminations have been compared with a function that is computed from the labeling data on the extreme assumption that the listener cannot hear any differences among these sounds beyond those that are revealed by his use of the phonemic labels."*

The authors are quite clear that while discrimination performance is substantially higher across a category boundary than within one, they also reject the extreme-CP view as a psychological theory, using it only for quantitative comparison: *"To obtain a more nearly precise evaluation of this effect [better discrimination at phoneme boundaries than in the middle of the phoneme category], the obtained discrimination curves were compared with those that would be predicted from the phoneme identification curves on a most extreme assumption: namely, that S[ubjects] can only hear those differences that are revealed by the way in which he attached phonemic labels to the stimuli. The discrimination curves that were produced on this assumption predicted fairly well the occurrence and location of points of high and low discriminability in the obtained functions, but S[ubject]s tended in general to discriminate the stimuli somewhat better than expected from the extreme assumption"*

We believe that the present Discrete Sampling framework is consistent not only with the empirical data but also with the spirit of what was originally described within Liberman et al. (1957) — the risk of engaging in hermeneutics on a paper nearly 70 years after its original publishing not withstanding.

B Alternative perceptual encodings

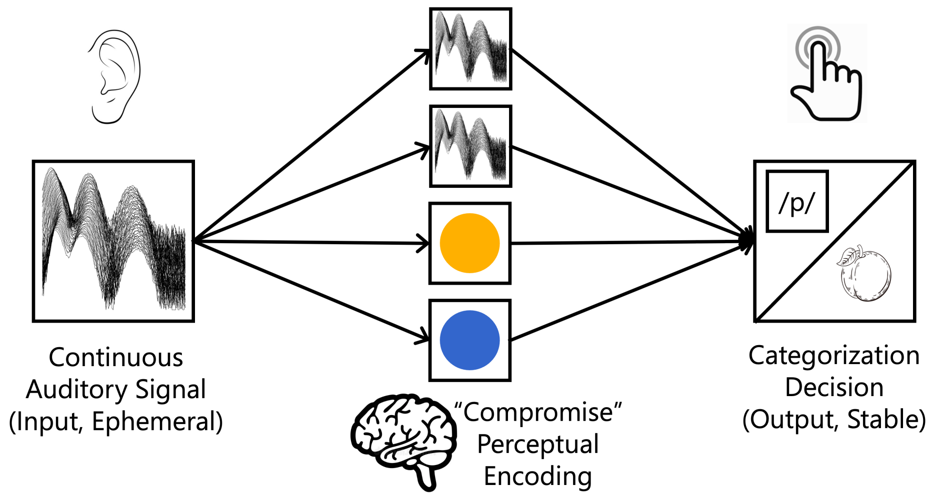

Figure B1: Schematic of a "compromise" theory of perceptual encoding; where the intermediate representation contains contains multiple classes of information simultaneously.

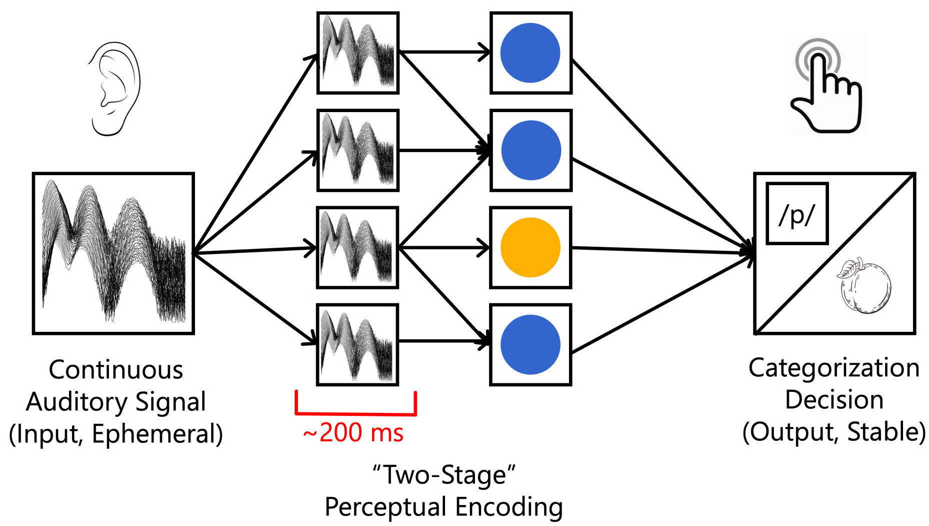

Figure B2: Schematic of a "two-stage" theory of perceptual encoding; here the auditory input is encoded as a continuous or cue-based representation at an early stage of initial processing (e.g. the first 200ms post-stimulus) before being converted into a discrete, category-based representation from about 200ms onward. The latter representation forming the basis of subsequent categorization and downstream decision making.

C Discrete Sampling Algorithm Specifications

C.1 Algorithm: input to encoding

```
Function input-to-encoding(input: float, mu: float, sigma: float, num_samples: int):
    labels = [] // Initialize an empty list for labels
    for i = 0 to num_samples - 1 do
        curr_thresh = random.normal(mu, sigma)
        if input < curr_thresh then
            labels.append('d') // Label as voiced '/d/'
        else
            labels.append('t') // Label as voiceless '/t/'
        end
    end
    n_d = count(labels, 'd') // Count occurrences
    n_t = count(labels, 't')
    e = {'d': n_d, 't': n_t} // final encoding
    return e
```

Algorithm 1: Algorithmic procedure to sample a single perceptual encoding e given some phonetic input in the Discrete Sampling (DS) model. $input$ is a primary acoustic cue such as VOT which is sampled $N = num\_samples$ times. mu is the general category boundary subject to $sigma$ variability.

C.2 Algorithm: encoding to category judgment

```
Function encoding-to-judgment(encoding: list, M: int):
    // Initialize with first draw
    prev_label = sample(encoding)
    curr_label = sample(encoding)
    total_draws = 1 // Track total calculations
    in_a_row = 1 // Consecutive matches
    while less than M consecutive identical labels drawn do
        total_draws += 1
        if curr_label == prev_label then
            in_a_row += 1
            if in_a_row == M then
                return (curr_label, total_draws) // Judgment and RT
            end
        else
            prev_label = curr_label
            in_a_row = 1
            curr_label = sample(encoding)
        end
    end
```

Algorithm 2: Algorithmic procedure to generate a single category judgment from an intermediate perceptual encoding under the Discrete Sampling (DS) model. $encoding$ is a list (multiset) of N labels, and M is the number of consecutive identical labels required to converge to a response. The number of draws until convergence is a natural measure of reaction time (RT).

C.3 Algorithm: encoding to scalar response

```
Function encoding-to-scalar(encoding: list, M: int):
    conv_threshold = 0.03 // threshold for convergence

    // initialize empty buffer and counters
    t_count, total_draws, t_count_M_ago = 0
    buffer = []

    // fill initial buffer
    for i = 0 to M do
        is_t = (sample(encoding) == 't')
        buffer.push(is_t)
        total_draws += 1
        if curr_label == prev_label then
            t_count += is_t
        end
    end
    t_count_M_ago = buffer.pop()

    while delta > conv_threshold do
        prev_ratio = t_count_N / (total_draws - M)
        is_t = (sample(encoding) == 't')
        t_count += is_t
        total_draws += 1
        t_count_M_ago = buffer.pop()
        buffer.push(is_t)

        curr_ratio = t_count / total_draws
        delta = curr_ratio - prev_ratio
    end

    return (curr_ratio, total_draws) // Scalar output and RT
```

Algorithm 3: Algorithmic procedure to generate a continuous scalar response category judgment for /t/-likeness from an intermediate perceptual encoding under the Discrete Sampling (DS) model. *encoding* is a list (multiset) of N labels, and M is the size of the buffer over which to calculate scalar convergence, with the number of draws until convergence as a natural measure of reaction time (RT).

D Glossary

Auditory input

The continuous acoustic signal reaching a listener's auditory system. Within DS, this input provides the raw, time-varying evidence from which discrete perceptual samples may be drawn

Primary cue

An acoustic dimension that is most diagnostic for a given phonetic contrast (e.g., voice-onset time in the case of for /t/-/d/). Although multiple cues may be present for any given auditory input or contrast, the primary cue is the one to which the perceptual system (descriptively) attributes the greatest weight to in the context of the task. For illustrative simplicity we reduce perception here to operating over a single primary cue per contrast.

Perceptual encoding

This is the intermediate mental representation constructed from the original, ephemeral auditory input. Although theories differ substantially in the content (cue-based vs. category-based) and granularity of the perceptual encoding, its presence is a functional necessity for any model of speech perception. It denotes a representation that can be maintained in short-term memory — irrespective of content or form — outlasting the original input but remaining available for integration with subsequent information or for decision processes such as phonological or lexical categorization. Within the Discrete Sampling framework, a multiset of labeled samples serves as the data structure by which this encoding is instantiated, though such implementation specifics are conceptually separable from the notion of a perceptual encoding in general.

Task-relevant response

An overt behavioral output in an experimental paradigm (e.g., a label, an identification choice, a slider position) derived via a task-specific decision rule from the perceptual encoding. The structure of the encoding need not map one-to-one with the response.

Sample

This is the atomic unit of representation for perceptual encodings in the Discrete Sampling framework: a single, category label generated on the basis of comparison between phonetic input and the category boundary μ, subject to noise σ (i.e. $\sim \mathcal{N}(\mu, \sigma^2)$) An ensemble of multiple samples generated within a short temporal window collectively constitutes the (multiset) perceptual encoding used to generate downstream judgments and inference.